Predicting Cluster Formation in Deep Graph Attention Networks: Phase Transitions, Topology, and Feature Geometry

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Abstract

Deep graph attention networks with a fixed temperature oversmooth: all node features collapse to one point. We ask the question that remains when depth does *not* collapse everything: which clusters survive, and can they be predicted from the input? We study Gaussian (distance-based) graph attention under logarithmic cooling $T_t = c/\log(t + t_0)$, the schedule at which merging and separation balance. Classical annealing arguments give the leading-order picture: feature gaps shorter than $\sqrt{c}$ close, longer gaps may survive. Our results concern what that picture misses. (i) For two blocks with arbitrary cut ratios we prove an exact threshold law. Gaps below $\sqrt{c}\,\kappa(t_0)$ merge and gaps at or above it persist. The limiting gap jumps from 0 to exactly $\sqrt{c}$ at the threshold. The threshold follows its own trajectory through depth. Topology enters only at second order, through the sum λ of the cut ratios: $\kappa(t_0) = 1 + [\log L - \frac{1}{2}\log\log L + \frac{1}{2}\log\lambda + o(1)]/L$ with $L = \log t_0$. (ii) In layered chains, neighbours can rescue a gap but never destroy one, so the isolated-pair rule makes only one-sided errors. The rescue is large: in three-layer chains the normalised rescue boundary converges to $\varphi(x) = \max\{x, 1 - x/2\}$, so gaps down to 2/3 of the threshold survive. (iii) On general graphs every limiting edge length is 0 or at least $\sqrt{c}$, and planted partitions survive at the $\sqrt{c}$ scale, with a matching impossibility result. (iv) For learned query–key attention, including dot-product attention with biases, the two-block law holds for the smaller of the two directed attention margins: exactly for symmetric pairs, and to leading order in general. At a fixed symmetric part, key biases and key norms only make clusters merge earlier, and a negative symmetric part makes features oscillate forever. Validated numerics certify a three-layer phase diagram (99.6% of a 240×240 grid). The theory gives a cheap diagnostic: threshold the margins after a few layers. It predicts 99.1% of final gaps in random chains after 10 layers and the final partition of noisy community graphs with ARI ≥ 0.99 for feature noise up to $\sigma = 0.3$. A learned metric W enters only through $z = Wh$, so the theory states which class distinctions a trained metric must place above the threshold. In trained dot-product networks that are near their limit the rule predicts the partition they converge to (ARI 0.97); in the others it shows that the accuracy at the trained depth rests on a transient that further depth destroys. We do not propose a new architecture: trained cooled Gaussian networks stay within 0.007 of an MLP at every depth, but residual and normalised baselines beat them where propagation helps.

1 Introduction

Stacking message-passing layers makes node features similar. For graph convolutions this oversmoothing is exponential in depth [4, 21, 26, 29], and Wu et al. [39] showed that attention with a fixed temperature cannot prevent it: features still converge to a single point. Self-attention on the sphere behaves differently. There, tokens form several clusters, which persist for exponentially long times [3, 11, 12]. Clusters, not consensus, are therefore the generic long-depth outcome of attention-type dynamics once the attention sharpens. This raises a question that the oversmoothing literature does not address: *given the graph and the input*

features, which clusters will a deep attention network form?

We answer this question for the simplest model in which it is non-trivial: Gaussian graph attention $\alpha_{ij} \propto \exp(-\|W(h_i - h_j)\|^2/T_t)$ with logarithmic cooling $T_t = c/\log(t + t_0)$. Under this schedule the unnormalised weight across a gap d decays like $t^{-d^2/c}$, which is summable exactly when $d > \sqrt{c}$. This is the mechanism of Hajek's criterion for simulated annealing [10, 13], and it gives the leading-order rule "gaps below $\sqrt{c}$ close". We do not claim this rule. Our point of departure is the observation that it is wrong at every finite depth, and wrong in a structured way. In one sentence: *logarithmic cooling turns a globally supported Gaussian kernel into an effective bounded-confidence model* in the sense of Hegselmann–Krause [1, 15], *whose confidence radius* $\sqrt{c}\,\kappa(t_0)$ *we compute exactly, and whose nonlocal corrections we quantify.* Learned query–key attention obeys the same law, applied to the smaller of the two directed attention margins of a pair (Section 7).

Contributions.

- **An exact two-block threshold law** (Theorem 1). For two blocks with arbitrary cut ratios ρ_A, ρ_B there is a threshold $\kappa(t_0; \rho_A, \rho_B) > 1$ such that gaps below $\sqrt{c}\,\kappa$ merge and gaps at or above it persist. At the threshold the limiting gap jumps from 0 to exactly $\sqrt{c}$, a first-order transition. The threshold obeys a critical-curve identity $\kappa(t_0 + t) = F_{t-1} \circ \cdots \circ F_0(\kappa(t_0))$. We determine its asymptotics to second order, $\kappa - 1 = [\log L - \frac{1}{2}\log\log L + \frac{1}{2}\log(\rho_A + \rho_B) + o(1)]/L$ with $L = \log t_0$: topology enters only at second order, and only through $\rho_A + \rho_B$. We also prove the third term, and give explicit two-sided bounds and validated enclosures.

- **One-sided errors and an early-depth rule** (Lemma 2 and Theorem 2). In layered chains, a gap can only gain from its neighbours. The isolated-pair rule therefore never predicts persistence wrongly, and the rule "persist iff $g(t_1) \geq \kappa(t_0 + t_1)$" becomes monotonically more accurate with t_1.

- **A quantitative nonlocal rescue law** (Theorems 3 and 4). For three layers we prove an exact persistence criterion, a rescue for every t_0 and every neighbouring gap, and the scaling limit $u^*/\kappa \to \varphi(v_0/\kappa)$ with $\varphi(x) = \max\{x, 1 - x/2\}$ for $x < 1$. Gaps down to $2/3$ of the threshold survive. A computer-assisted theorem certifies the finite-t_0 phase diagram (Figure 1).

- **General graphs** (Propositions 3 and 4 and Theorem 5). Every limiting edge length is 0 or $\geq \sqrt{c}$. Planted partitions survive when blocks have diameter below $\sqrt{c}$ and adjacent blocks are separated by $\kappa(t_0)\sqrt{c}$ to leading order, and the diameter condition cannot be relaxed.

- **Learned attention** (Section 7). For quadratic scores, which include dot-product attention with query and key biases, the margin of an edge splits into a Gaussian part, a key potential and a rotation (Proposition 5). On two blocks the dynamics depends on the score only through the two directed margins. The pair merges iff the smaller margin falls to c. Symmetric pairs follow the Gaussian law exactly, and in general the critical smaller margin is $c\,\kappa(t_0)^2$ to leading order (Theorem 6). A score with a negative symmetric part makes features oscillate forever (Proposition 6), and GAT scores cannot hold two clusters apart in their linear regime.

- **Prediction in practice** (Section 8). The early-depth rule predicts 99.1% of final gaps after 10 layers in random five-layer chains. On noisy contextual SBMs the threshold-graph rule recovers the final partition with ARI ≥ 0.99 up to $\sigma = 0.3$, including fragmentation into ≈ 7 clusters. With a learned metric, the theory states which class distinctions survive depth.

- **Trained networks** (Section 9). We train cooled Gaussian and dot-product networks against ten baselines on five synthetic tasks and two real graphs, and rerun 150 trained

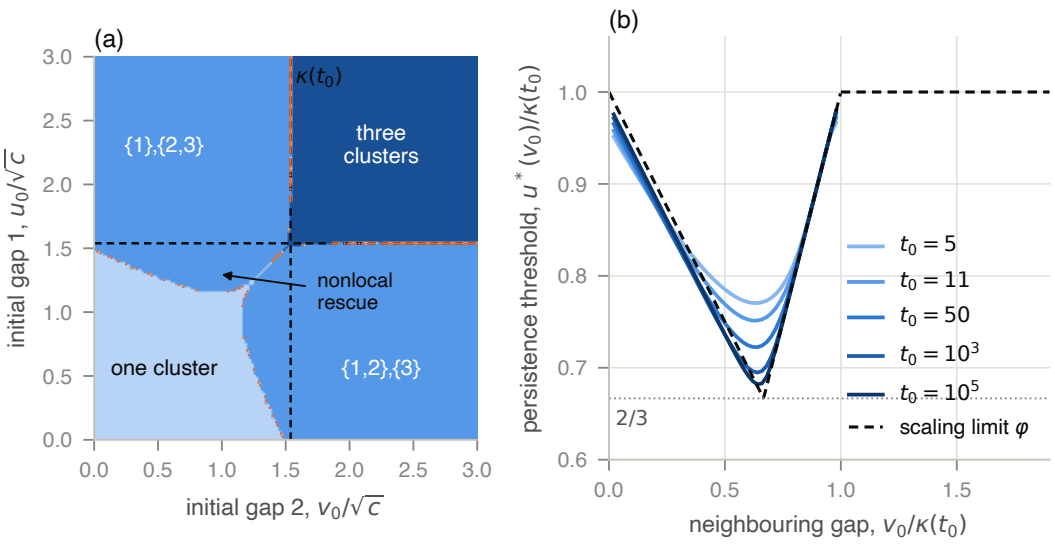


Figure 1: **Nonlocal rescue in the three-layer chain.** (a) Certified limiting partition at $t_0 = 11$ on a 240×240 grid of initial gaps (u_0, v_0), in units of $\sqrt{c}$. Shades give the number of final clusters. Orange dots mark the 208 grid points (0.36%) that the validated computation did not resolve; they lie on phase boundaries. Dashed lines mark the isolated-pair threshold $\kappa(11) \approx 1.539$. Below the horizontal line, gap 1 would merge if it were isolated. In the bulge it persists, because gap 2 closes first and pulls layer 2 away (Theorem 3). (b) Normalised persistence threshold $u^*(v_0)/\kappa(t_0)$ for t_0 from 5 to 10^5 (light to dark; float estimates), shown for $v_0 < \kappa(t_0)$, together with the scaling limit φ of Theorem 3(c). The minimum approaches 2/3.

dot-product networks to 4000 layers. A trained network is either near its limiting partition, which the directed rule predicts after a few layers, or in a transient whose accuracy further depth destroys.

Scope. We do not propose cooled attention as a new architecture. Cooled Gaussian attention is the analysable model: trained, it never falls more than 0.005 below an MLP in accuracy at any depth on our seven node tasks (0.007 in average precision on Peptides-func), because the frozen phase is always reachable, but residual and normalised baselines beat it wherever propagation helps. The payoff is predictive: which distinctions a deep attention network keeps, which metric keeps them, and whether a trained network's accuracy at its trained depth will survive further depth.

2 Related work

Oversmoothing and attention. Oversmoothing of graph convolutions is exponential in depth [4, 21, 26]; see Rusch et al. [29] for a survey. Wu et al. [39] proved that graph attention with a fixed temperature also oversmooths. Ma et al. [22] characterise when attention helps over convolution (structure noise versus feature noise); our noisy-feature regime, in which PairNorm is the best model from depth 32 on, agrees with their prediction that attention does not help when feature noise dominates. Turan et al. [34] obtain non-homogeneous limits from activation-induced bifurcations, a different mechanism. Zhang et al. [40] argue that oversmoothing should be measured by the effective rank [28]; we report it. Remedies include residual and initial connections [6, 20], normalisation [41] and continuous diffusion [5]; we use them as baselines in Section 9.

Clustering in attention dynamics. Geshkovski et al. [11] showed that self-attention dynamics on the sphere cluster. Clusters are metastable and merge after exponentially long times [3, 12]; causal masking changes the limit [18]. Gao et al. [9] report an attention-condensation transition above a sharpness threshold. These works study transformers at a fixed inverse temperature without a graph. Our cooling schedule turns metastable clusters into permanent ones, and the graph supplies the topology whose effect we quantify. Brody et al. [2] showed that GAT computes static attention, in which the ranking of keys does not depend on the query; Remark 6 gives a dynamical consequence.

Bounded confidence and consensus. In the Hegselmann–Krause model [15] agents average only over neighbours within a confidence radius R. Limiting clusters are at least R apart [1], compactly supported kernels give local consensus [17, 25], and whether two groups merge can depend on agents between them: mediator agents, equally spaced agents and the $2R$ conjecture [14, 30, 36]. Our Proposition 3 is the soft-kernel, cooled analogue of the separation fact. Our rescue law is a quantitative counterpart of the qualitative mediator phenomenon: we did not find an exact rescue boundary in this literature. Merging along edges with non-summable weights is the infinite-flow theorem of Touri and Nedić [32]. The cut-balanced theorem of Hendrickx and Tsitsiklis [16] does not apply in discrete time here, because the attention weights are not bounded below.

Annealing. For simulated annealing with $T_t = c/\log t$, Hajek [13] showed convergence to global minima iff c is at least the depth of the deepest non-global well, through the same summability dichotomy. The leading-order $\sqrt{c}$ threshold is therefore classical. The exact finite-t_0 law, the jump, the critical curve and the topology correction are not covered by these results.

Bounded confidence and temperature as architectures. Hegselmann–Krause-style message passing has been proposed to mitigate oversmoothing [37, 42], and learnable attention temperatures exist [23]. We do not propose cooled Gaussian attention as a new architecture. We use it as the analysable model in which cluster formation can be predicted, and we test in Section 9 where it helps and where it does not.

3 Setup

Model. Let $G = (V, E)$ be a finite undirected graph with a self-loop at every node, $N[i]$ the closed neighbourhood of i and $N_{\max} = \max_i |N[i]|$. Node features $h_i(t) \in \mathbb{R}^p$ evolve by

$$h_i(t+1) = \sum_{j \in N[i]} \alpha_{ij}(t)\, h_j(t), \qquad \alpha_{ij}(t) = \frac{\exp\big(-\|W(h_i(t) - h_j(t))\|^2/T_t\big)}{\sum_{k \in N[i]} \exp\big(-\|W(h_i(t) - h_k(t))\|^2/T_t\big)}. \quad (1)$$

This is the propagation core of a deep attention GNN: one shared scoring rule, no value matrix and no nonlinearity, as in oversmoothing analyses [11, 39]. The metric $W \in \mathbb{R}^{q \times p}$ is fixed (learned, in Section 8).

Remark 1 (the metric is absorbed). Let $z_i = Wh_i$. Then $z_i(t+1) = \sum_j \alpha_{ij}(t) z_j(t)$ with α computed from $\|z_i - z_j\|$, so z follows (1) with $W = I$, and every result below applies to z. The values h use the same weights. By Proposition 3, edges whose z-endpoints merge have non-summable weights, so their h-endpoints merge as well. Edges whose z-endpoints stay apart have $W(h_i^\infty - h_j^\infty) \neq 0$. Hence *the limiting partition of h equals that of z*, and a learned metric decides which distinctions survive only through the distances $\|W(h_i - h_j)\|$.

Temperature. With a constant temperature the dynamics oversmooths.

Proposition 1 (constant temperature)**.** *If G is connected and $T_t \equiv T > 0$, then all $h_i(t)$ converge to a common limit, exponentially fast.*

We therefore study logarithmic cooling $T_t = c/\log(t + t_0)$ with $c > 0$ and $t_0 > 1$ (our graph experiments use $t_0 = |V| + 2$). We measure lengths in units of $\sqrt{c}$ and write $s_t := t + t_0$. With $W = I$ the unnormalised weights become

$$w_{ij}(t) = s_t^{-\|z_i(t) - z_j(t)\|^2}, \qquad \alpha_{ij} = w_{ij}/\pi_i, \qquad \pi_i = \sum_{k \in N[i]} w_{ik} \in [1, N_{\max}]. \tag{2}$$

Since $\sum_t s_t^{-d^2}$ diverges iff $d \le 1$, a gap that stays below 1 is forced to close, and a gap that stays above 1 sees summable attention. The difficulty is that gaps move. An edge *merges* if $\|z_i - z_j\| \to 0$ and *persists* if its length has a positive $\liminf$.

Two-block graphs. A graph with vertex partition $V = A \sqcup B$ is *block-regular* if every node of A has exactly k_A closed neighbours in A and $e_A \ge 1$ neighbours in B, and symmetrically for B. Its *cut ratios* are $\rho_A = e_A/k_A$ and $\rho_B = e_B/k_B$. We write $\lambda = \rho_A + \rho_B$, $P = \rho_A \rho_B$, and assume $P \le 1$. Examples: two layers of the layered graph below ($\rho_A = \rho_B = 1$); two k-cliques joined by a perfect matching ($\rho = 1/k$).

Lemma 1 (two-block reduction)**.** *On a block-regular graph, block-constant features stay block-constant, and the gap $d_t = \|z_A(t) - z_B(t)\|$ obeys*

$$d_{t+1} = F_t(d_t) := d_t\, \psi(a_t), \qquad \psi(a) = \frac{1 - Pa^2}{(1 + \rho_A a)(1 + \rho_B a)}, \qquad a_t = s_t^{-d_t^2}. \tag{3}$$

For $\rho_A = \rho_B = 1$ this is $d' = d\,(1 - a)/(1 + a)$.

Layered chains. The layered graph $G_{r,m}$ has r layers of m nodes. Each layer is a clique, consecutive layers are completely joined, and there are no other edges. With scalar, layer-constant, ordered features $x_1 \le \cdots \le x_r$, the gaps $g_l = x_{l+1} - x_l$ $(1 \le l < r)$ obey, with $w_l = s_t^{-g_l^2}$, $w_0 = w_r = 0$, $D_l = 1 + w_{l-1} + w_l$, $A = 1 + w_{l-1}$ and $B = 1 + w_{l+1}$,

$$g_l' = g_l\, \frac{AB - w_l^2}{D_l D_{l+1}} + \frac{w_{l+1} g_{l+1}}{D_{l+1}} + \frac{w_{l-1} g_{l-1}}{D_l}. \tag{4}$$

All three terms are nonnegative. The first is the isolated two-layer map with extra damping; the other two are pushes from the neighbouring gaps.

Validated numerics. Several results are computer-assisted. We evaluate (3) and (4) in interval arithmetic [33] and stop as soon as an analytic certificate proves the fate of every gap (Section E). Persistence is certified by a tail bound (certificate C1, Theorem 1(vi)), merging by a block-diameter bound (C2). Statements called *proved* use `mpmath` interval arithmetic at 160 bits [31]. Statements called *certified* use float64 intervals with outward rounding, where exp and log are padded by 4 ulp; they are rigorous provided the platform's elementary functions are accurate to 4 ulp.

4 Two blocks: an exact threshold law

Theorem 1 (two-block law)**.** *Let $\rho_A, \rho_B > 0$ with $P = \rho_A \rho_B \le 1$, $t_0 > 1$, and let d_t follow (3) from $d_0 > 0$.*

(i) Dichotomy. *$d_\infty = \lim_t d_t$ exists and $d_\infty \in \{0\} \cup [1, \infty)$. Moreover $d_\infty = 0$ iff $d_T \le 1$ for some T.*

(ii) Threshold. *There is* $\kappa = \kappa(t_0; \rho_A, \rho_B) \in (1, \infty)$ *such that the gap merges iff* $d_0 < \kappa$.

(iii) First-order jump. $d_0 \mapsto d_\infty$ *is nondecreasing and right-continuous, and* $d_\infty(\kappa) = 1$. *The limiting gap jumps from* 0 *to exactly* $\sqrt{c}$ *at the threshold.*

(iv) Critical curve and monotonicity. $\kappa(t_0 + k) = F_{k-1} \circ \cdots \circ F_0(\kappa(t_0))$ *for* $k \in \mathbb{N}$. *Moreover* κ *is nonincreasing in* t_0 *and nondecreasing in* ρ_A *and in* ρ_B.

(v) Second-order asymptotics. *For fixed* ρ_A, ρ_B, *with* $L = \log t_0 \to \infty$,

$$\kappa(t_0; \rho_A, \rho_B) = 1 + \frac{\log L - \frac{1}{2}\log\log L + \frac{1}{2}\log(\rho_A + \rho_B) + o(1)}{L}.$$

Equivalently, in physical units with initial temperature $T_0 = c/\log t_0$, *the critical gap* d_{crit} *satisfies* $d_{\mathrm{crit}}^2 = c + T_0\big[2\log\frac{c}{T_0} - \log\log\frac{c}{T_0} + \log(\rho_A + \rho_B) + o(1)\big]$ *as* $T_0 \to 0$.

(vi) Explicit bounds. $\kappa(t_0) \geq 1 + \varepsilon_-$, *where* ε_- *is the unique root of* $\varepsilon(2\varepsilon + \varepsilon^2) = \lambda_- \, t_0^{-(2\varepsilon+\varepsilon^2)}$ *and* $\lambda_- = \sum_{X \in \{A,B\}} \rho_X/(1 + \rho_X/t_0)$. *Conversely (certificate C1), if* $1 < \delta < d_0$ *and*

$$\log d_0 - \Lambda_\delta \, \frac{(t_0 - 1)^{1-\delta^2}}{\delta^2 - 1} \;\geq\; \log\delta, \qquad \Lambda_\delta = \lambda + \frac{Pq}{1 - Pq^2}, \quad q = t_0^{-\delta^2}, \tag{5}$$

then $d_t \geq \delta$ *for all* t, *so* $\kappa(t_0) \leq d_0$.

The proof is in Section A. Part (i) is the soft-kernel analogue of the bounded-confidence separation fact [1], and the limit $\kappa \to 1$ in part (v) is Hajek's summability criterion [13]. The new content is (ii)–(vi) at finite t_0 and the expansion in (v), whose error is $o(1/\log t_0)$. The proof of (v) compares the trajectory with explicit barriers $1 + X(\log s_t)$, $X(\tau) = (\log\tau - \frac{1}{2}\log\log\tau + \frac{1}{2}\log\lambda \pm \eta)/\tau$: with $+\eta$ the map cannot cross the barrier from above, with $-\eta$ it cannot cross from below, and below the lower barrier the accumulated attention exceeds the available gap. In physical units the effective confidence radius is $\sqrt{c}\,\kappa(t_0)$, not $\sqrt{c}$. The excess $d_{\mathrm{crit}}^2 - c \approx 2T_0\log(c/T_0)$ is the distance that a gap loses before its own weight becomes summable. The correction is large in practice: $\kappa(11) \in [1.53858, 1.53909]$, so at $t_0 = 11$ a gap must be 54% above $\sqrt{c}$ to survive (Table 2). Figure 2 shows the enclosures, the jump and the critical slowing down implied by (iii) and (iv). Trajectories started just above κ follow the critical curve for up to 10^6 layers before they settle or merge.

Corollary 1 (topology enters at second order)**.** *For fixed cut ratios with sums* $\lambda = \rho_A + \rho_B$ *and* $\lambda' = \rho'_A + \rho'_B$,

$$\kappa(t_0; \rho_A, \rho_B) - \kappa(t_0; \rho'_A, \rho'_B) = \frac{\frac{1}{2}\log(\lambda/\lambda') + o(1)}{\log t_0}.$$

In physical units the critical squared gap shifts by $T_0\log(\lambda/\lambda') + o(T_0)$. *The product* $P = \rho_A\rho_B$ *does not enter at this order.*

The reading is an energy–entropy balance. A gap d shrinks at relative rate $\approx \lambda\, e^{-d^2/T_t}$, so multiplying the cut ratios by a factor is equivalent to shifting d^2 by T_t times its logarithm. Topology matters at finite depth: $\kappa(11)$ ranges from 1.30 for $\rho_A = \rho_B = 0.1$ to 1.54 for $\rho_A = \rho_B = 1$ (Table 3). The $o(1)$ term converges slowly when λ is small.

Proposition 2 (third term)**.** *For fixed* ρ_A, ρ_B, *with* $\lambda = \rho_A + \rho_B$ *and* $L = \log t_0 \to \infty$,

$$\kappa(t_0; \rho_A, \rho_B) = 1 + \frac{1}{L}\Big[\log L - \tfrac{1}{2}\log\log L + \tfrac{1}{2}\log\lambda + \frac{\log\log L - \log\lambda + 2 + o(1)}{4\log L}\Big].$$

In particular, the topology shift of Corollary 1 is $\frac{\log(\lambda/\lambda')}{2L}\big(1 - \frac{1}{2\log L}\big) + o\big(\frac{1}{L\log L}\big)$.

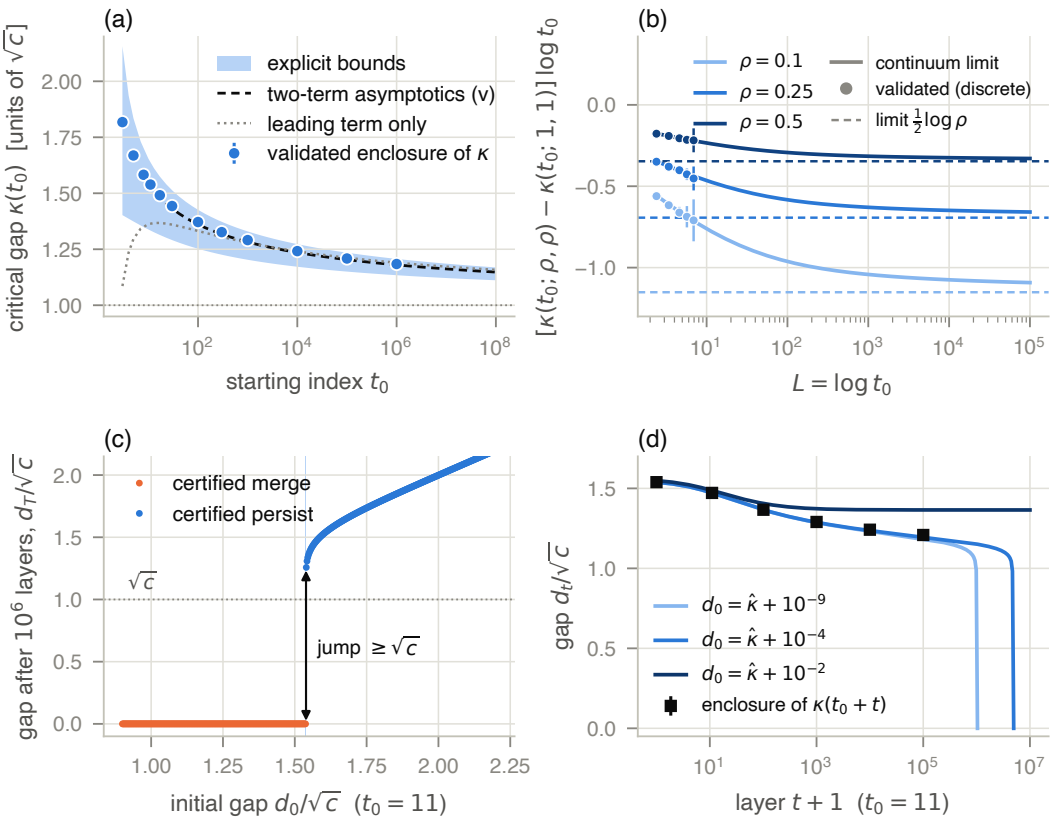


Figure 2: **The two-block law (Theorem 1).** (a) Validated enclosures of $\kappa(t_0)$ for $\rho_A = \rho_B = 1$ (dots; widths below 0.05), the explicit bounds of part (vi) (band), the two-term asymptotics of part (v) (dashed) and its leading term alone (dotted). (b) Topology at second order (Corollary 1): the threshold shift relative to $\rho = 1$, multiplied by $\log t_0$, tends to $\frac{1}{2}\log\rho$ (dashed). Dots: validated enclosures of the discrete map ($t_0 \le 10^3$); lines: continuum limit up to $t_0 = e^{10^5}$. The approach is slow, with relative error $\approx 1/(2\log\log t_0)$ (Proposition 2). (c) Gap after 10^6 layers against the initial gap at $t_0 = 11$. Every point is certified to merge (orange) or to persist (blue). The gap jumps from 0 to at least $\sqrt{c}$ at κ (shaded, width $5 \cdot 10^{-4}$). (d) Critical slowing down. Trajectories started just above the float estimate $\hat{\kappa}$ of $\kappa(11)$ follow the critical curve $\kappa(t_0 + t)$ (squares: validated enclosures, part (iv)) for up to 10^6 layers.

The proof (Section A.1) uses the barriers of (v) with c replaced by $\frac{1}{2}\log\lambda + (\log\log\tau - \log\lambda + 2 \pm \eta)/(4\log\tau)$. At this order the $\pm\eta$ term decides the sign of every comparison, including the summation that rules out persistence below the lower barrier. The topology shift therefore approaches its limit only like $1 - 1/(2\log\log t_0)$, which is why it converges so slowly in Figure 2(b).

Remark 2 (accuracy at finite t_0). For $\lambda = 2$ the two-term formula of (v) is already accurate at moderate t_0: it lies below the validated enclosures by at most $0.05/\log t_0$ for $17 \le t_0 \le 10^4$ (Table 2). For smaller λ the third term is larger and the two-term formula converges more slowly (Table 4). Adding the third term reduces the residual by an order of magnitude for $L \gtrsim 10^3$, while at small L it overcorrects. The root ε_- in (vi) has $-\log\log L$ instead of $-\frac{1}{2}\log\log L$ in its expansion: the bound freezes the gap, while the true gap shrinks along the critical curve, and this costs the extra $\frac{1}{2}\log\log L/L$.

5 Chains: one-sided errors and nonlocal rescue

Let F_t denote the isolated two-layer map ($\rho_A = \rho_B = 1$) and $\kappa(t_0) = \kappa(t_0; 1, 1)$.

Lemma 2 (comparison)**.** *Along* (4) *all gaps stay nonnegative and* $g_l(t+1) \geq F_t(g_l(t))$ *for every* l *and* t*.*

The neighbouring gaps can only push a gap open. This has three consequences.

Theorem 2 (one-sided errors)**.** *Consider* $G_{r,m}$ *with ordered layer-constant features.*

(a) If $g_l(t_1) \geq \kappa(t_0 + t_1)$ *for some* t_1*, then* $g_l(t) \geq 1$ *for all* $t \geq t_1$*; the gap persists.*

(b) The isolated-pair rule "gap l *persists iff* $g_l(0) \geq \kappa(t_0)$*" never predicts persistence for a gap that merges. All of its errors are* rescues*: gaps below* $\kappa(t_0)$ *that persist.*

(c) The early-depth rule $\mathcal{R}_{t_1}$*: "gap* l *persists iff* $g_l(t_1) \geq \kappa(t_0 + t_1)$*" has the same one-sided error. The set of gaps it declares persistent is nondecreasing in* t_1*. Every gap with* $\lim_t g_l(t) \neq 1$ *is classified correctly for all large* t_1*.*

The early-depth rule needs only t_1 layers of forward computation and the scalar function κ, which we tabulate once. Section 8 shows that it is accurate already at $t_1 = 10$.

Three layers. We now quantify the rescue. Write $u = g_1$, $v = g_2$ for the two gaps of $G_{3,m}$ and $h = u + v/2$, the distance from layer 1 to the midpoint of layers 2 and 3. The *rescue boundary* $u^*(v_0)$ is the smallest u_0 from which gap 1 persists; we define the upper and lower versions $U_\pm$ in Section B.1.

Theorem 3 (nonlocal rescue law)**.** *Let* $t_0 \geq 2$*.*

(a) Exact criterion. *Gap 1 merges whenever* $u_0 + v_0/2 < 1$*. If* $u_0 + v_0 < 2$*, gap 1 persists iff* $h(t) \geq 1$ *for all* t*; in that case layers 2 and 3 merge and gap 1 converges to* $\lim_t h(t)$*. On the symmetric line* $u_0 = v_0 < 1$ *both gaps merge.*

(b) Rescue at every t_0. *For every* $v_0 > 0$ *there is an explicit* $R(t_0, v_0) > 0$ *such that gap 1 persists for every* $u_0 \geq \kappa(t_0) - R$*. The isolated rule is wrong on* $[\kappa - R, \kappa)$*; the neighbour pushes* u *across the critical curve in one step.*

(c) Scaling limit. *Let* $\varphi(x) = \max\{x, 1 - x/2\}$ *for* $0 < x < 1$ *and* $\varphi(x) = 1$ *for* $x > 1$*. For every* $x \in (0, 1) \cup (1, \infty)$*,*

$$\lim_{t_0 \to \infty} \frac{U_\pm(x\,\kappa(t_0); t_0)}{\kappa(t_0)} = \varphi(x), \qquad \lim_{t_0 \to \infty} \inf_{v_0 > 0} \frac{U_\pm(v_0; t_0)}{\kappa(t_0)} = \varphi\big(\tfrac{2}{3}\big) = \tfrac{2}{3}.$$

The limit φ has a simple reading. If the neighbouring gap is small ($x < 2/3$), it closes first, and layer 1 then faces the merged pair at distance $h = u + v/2$: gap 1 survives iff h clears the threshold, giving $1 - x/2$. If $2/3 < x < 1$, the two gaps race and the larger one survives ($\varphi = x$). If $x > 1$, the neighbour persists itself and does not help ($\varphi = 1$). Rescue is strongest at $x = 2/3$, where a gap one third below the isolated threshold still survives. At finite t_0 the minimum is $0.770, 0.751, 0.722, 0.695, 0.682$ for $t_0 = 5, 11, 50, 10^3, 10^5$ (Figure 1b). The constant T^* beyond which (c) applies is explicit but huge near the corner $x = 2/3$, so for moderate t_0 we rely on computer-assisted proofs.

Theorem 4 (computer-assisted, $t_0 = 11$)**.** *(a) (proved)* $1.538 < \kappa(11) < 1.540$*.*

(b) (proved) The six initial conditions in Table 5 have the stated limiting partitions. In particular, at $u_0 = 1.4 < \kappa(11)$ *the fate of gap 1 is not monotone in* v_0*: it merges for* $v_0 = 0.1$*, persists for* $v_0 \in \{0.5, 0.95\}$ *and merges for* $v_0 \in \{1.5, 3.0\}$*. Also* $u_0 = 1.2$*,* 22% *below* $\kappa(11)$*, persists for* $v_0 = 0.95$*, and* $v_0 = 1.5 < \kappa(11)$ *persists for* $u_0 = 1.4$*.*

(c) (certified) On the grid $\{0.0125, 0.025, \ldots, 3\}^2$*,* 57,392 *of* 57,600 *initial conditions (99.64%) have a certified limiting partition (Figure 1a). The smallest certified persisting gap is* $u_0 = 1.1625$ *(*24% *below* $\kappa(11)$*), at* $v_0 = 0.8875$*. No column of the grid has a non-monotone fate in* u_0*.*

6 General graphs: dichotomy and planted partitions

We now return to (1) on an arbitrary graph with vector features ($W = I$ by Remark 1).

Proposition 3 (universal dichotomy)**.** *For every graph with self-loops, every $t_0 > 1$ and every initial condition: (i) every $z_i(t)$ converges to some z_i^∞; (ii) if $\sum_t w_{ij}(t) = \infty$ on an edge $\{i, j\}$, then $z_i^\infty = z_j^\infty$; (iii) every limiting edge length $\|z_i^\infty - z_j^\infty\|$ lies in $\{0\} \cup [1, \infty)$.*

Limiting edge lengths in $(0, \sqrt{c})$ are therefore impossible on every graph. The limiting partition is well defined: its blocks are the connected components of the edges of length 0. The proof applies the infinite-flow theorem of Touri and Nedić [32] to the attention matrices, which have a uniformly positive diagonal and are balanced in their sense. Next, let $V = B_1 \sqcup \cdots \sqcup B_K$ be a planted partition. Blocks $k \neq l$ are *adjacent* ($k \sim l$) if an edge joins them, $e_{\max}$ is the largest number of neighbours a node has outside its block, and $C_k \supseteq \{z_i(0) : i \in B_k\}$ is convex with diameter D_k.

Theorem 5 (planted partitions at the $\sqrt{c}$ scale)**.** *Let $t_0 \geq 2$, let each $G[B_k]$ be connected, and let $\eta \geq 0$ and $\delta > 1$ satisfy*

$$\textit{(A1)}\quad D_k + 2\eta \leq 1 \quad \forall k, \qquad \textit{(A2)}\quad \operatorname{dist}(C_k, C_l) \geq \delta + 2\eta \quad \forall k \sim l,$$

$$\textit{(A3)}\quad e_{\max}\, \delta\, \frac{(t_0 - 1)^{1-\delta^2}}{\delta^2 - 1} \leq \eta.$$

Then $z_i(t) \in C_k + \bar{B}(\eta)$ for all t and $i \in B_k$. Each block converges to a single point z_k^∞, adjacent blocks end at distance $\geq \delta$, and the limiting partition is exactly $\{B_1, \ldots, B_K\}$.

Corollary 2 (ball form and leading-order sharpness)**.** *(a) If block k lies in a ball of radius r_k around μ_k, it suffices that $r_k \leq \frac{1}{2} - \eta$ and $\|\mu_k - \mu_l\| \geq r_k + r_l + \delta + 2\eta$ for $k \sim l$, with (A3). In physical units: $r \leq \sqrt{c}\,(\frac{1}{2} - \eta)$ and $\Delta \geq 2r + (\delta + 2\eta)\sqrt{c}$. (b) Fix $e_{\max}$ and $\theta \in (0, 1]$. If $D_k \leq 1 - \theta$ and $\operatorname{dist}(C_k, C_l) \geq \Gamma(t_0)$ for $k \sim l$, with an explicit $\Gamma(t_0) = 1 + (1 + o(1))\frac{\log\log t_0}{\log t_0}$, the conclusions hold for all large t_0. By Theorem 1(v), Γ matches the two-block threshold $\kappa(t_0; \rho_A, \rho_B)$ to leading order for every topology, and separation $\kappa(t_0; \rho_A, \rho_B)$ is necessary for block-regular graphs with collapsed blocks. At second order Γ is not sharp: $\Gamma - \kappa = (\frac{1}{2} + o(1)) \log\log\log t_0 / \log t_0$.*

Proposition 4 (the diameter condition is sharp)**.** *For every $D > 1$ and $t_0 \geq 2$ there is a graph with a single connected block and initial features of diameter D whose limit is not a consensus. Hence no condition $\max_k D_k \leq f(t_0)$ with $f(t_0) > 1$ guarantees the planted partition on all graphs.*

The construction is two large cliques joined by a perfect matching: the cut ratio $1/k$ makes the threshold of Theorem 1 drop towards 1. Theorem 5 removes the factor $1/(4\sqrt{\rho})$ (ρ the hop radius of a block) that our earlier path-product argument required. On the noisy SBMs of Section 8 it certifies all 100 runs with $\sigma \leq 0.1$, against 28 runs for the earlier condition. It never certifies a run in which the planted partition fails.

7 Learned query–key attention

So far the score has been a squared distance. Practical attention layers score pairs with learned queries and keys. We now replace the Gaussian score by a quadratic score with the same cooling,

$$\alpha_{ij}(t) \propto \exp\big(f(z_i, z_j) \log s_t\big) \quad (j \in N[i]), \qquad f(x, y) = x^\top J y - \tfrac{1}{2} y^\top \Omega y + b^\top y, \tag{6}$$

with $c = 1$. Dot-product attention with query and key biases, $(W_Q x + q)^\top (W_K y + k)$, is the case $J = W_Q^\top W_K$, $\Omega = 0$ and $b = W_K^\top q$; terms that depend only on the query x cancel in the

softmax. Gaussian attention with metric W is the case $J = \Omega = 2W^\top W$, $b = 0$. Because $N[i]$ contains i, the weights depend on f only through the *directed margins*

$$m_{ij} = f(z_i, z_i) - f(z_i, z_j), \qquad w_{ij} = s_t^{-m_{ij}}, \qquad \alpha_{ij} = w_{ij}/\pi_i, \qquad \pi_i = \textstyle\sum_{k\in N[i]} w_{ik} \geq 1.$$

The margin m_{ij} is how much node i prefers itself to j. The Gaussian margin $\|z_i - z_j\|^2$ is nonnegative and symmetric; a learned margin is neither.

Proposition 5 (margin decomposition)**.** *Let J_s and J_a be the symmetric and antisymmetric parts of J, and $\mathcal{V}(x) = \frac{1}{2}x^\top(J_s - \Omega)x + b^\top x$. Then*

$$m_{ij} = \tfrac{1}{2}(z_i - z_j)^\top J_s(z_i - z_j) + \mathcal{V}(z_i) - \mathcal{V}(z_j) - z_i^\top J_a z_j, \qquad m_{ij} + m_{ji} = (z_i - z_j)^\top J_s(z_i - z_j).$$

The attention weight is thus a Gaussian kernel in the metric $J_s/2$, times a key prior $s_t^{\mathcal{V}(z_j)}$, times a rotation factor. The Gaussian model is the case in which $\mathcal{V}$ is constant and $J_a = 0$. In dot-product attention $\mathcal{V}(x) = \frac{1}{2}x^\top J_s x + b^\top x$, so nodes with large keys attract their neighbours. The sum of the two margins of an edge is fixed by J_s alone. The potential and the rotation only split it between the two directions.

Two blocks. On a block-regular graph, block-constant features stay block-constant for every score, and

$$z'_B - z'_A = k\,(z_B - z_A), \qquad k = 1 - p_A - p_B = \frac{1 - P a_A a_B}{(1 + \rho_A a_A)(1 + \rho_B a_B)}, \qquad p_X = \frac{\rho_X a_X}{1 + \rho_X a_X},$$

with $a_A = s_t^{-m_{AB}}$ and $a_B = s_t^{-m_{BA}}$ (Lemma 9); for $a_A = a_B$ this is the factor ψ of Lemma 1. The gap keeps its direction e, up to sign. Write $M_A = m_{AB}$, $M_B = m_{BA}$, $\sigma = e^\top J_s e$ and $\omega = e^\top \Omega e/\sigma$. For the score (6) with $\sigma > 0$ the two margins form a closed system,

$$M'_X = k\Big[M_X - (M_A + M_B)\big(\tfrac{\omega}{2}(p_A + p_B) + (1 - \omega)\,p_X\big)\Big], \qquad X \in \{A, B\}, \tag{7}$$

with $M_A + M_B = \sigma\|z_A - z_B\|^2$. Since $a_A a_B = s_t^{-(M_A + M_B)} < 1$, $k > 0$: the blocks never cross. The fate of the pair depends on J, Ω and b only through the two margins and ω: dot-product attention has $\omega = 0$, Gaussian attention $\omega = 1$.

Theorem 6 (two blocks under learned attention)**.** *Let $\sigma > 0$, $0 \leq \omega \leq 1$ and $P = \rho_A\rho_B \leq 1$.*

(i) Dichotomy. *The blocks merge iff $\min(M_A, M_B) \leq 1$ at some layer. Otherwise both margins decrease to limits ≥ 1, and the blocks converge to two distinct points.*

(ii) Symmetric pairs. *If $\rho_A = \rho_B$ and $M_A = M_B$ at $t = 0$, then $M_A = M_B$ for all t, and $\sqrt{M_A}$ follows the Gaussian map F_t of (3). Hence the pair persists iff $M_A(0) \geq \kappa(t_0; \rho_A, \rho_A)^2$, and Theorem 1 applies to $\sqrt{M_A}$.*

(iii) Bounds. *If the pair persists, then $\min(M_A, M_B) \geq 1 + \varepsilon_\omega$ at $t = 0$, where ε_ω is the root of $\varepsilon^2 = (1 - \frac{\omega}{2})\rho_- t_0^{-\varepsilon}$ and $\rho_- = \min_X \rho_X/(1 + \rho_X/t_0)$. Conversely (certificate C1′), let $1 < \delta^2 < m := \min(M_A, M_B)$, $R = (M_A + M_B)/\delta^2$, $q = t_0^{-\delta^2}$ and $R\lambda q < 1$. If*

$$\log\frac{m}{\delta^2} \;\geq\; \frac{(1 + R)\lambda}{1 - R\lambda q} \cdot \frac{(t_0 - 1)^{1-\delta^2}}{\delta^2 - 1},$$

then both margins stay above δ^2 and the pair persists.

(iv) Leading order. *For every $\bar{M}$ there are $\gamma_\pm(t_0) = (2 + o(1))\log\log t_0/\log t_0$ such that a pair with $M_A + M_B \leq \bar{M}$ merges if $\min(M_A, M_B) < 1 + \gamma_-$ and persists if $\min(M_A, M_B) \geq 1 + \gamma_+$. To leading order the critical margin is $\kappa(t_0)^2$.*

The isolated-pair rule of Section 4 therefore carries over with one change: the squared gap is replaced by the smaller of the two directed margins. A pair is lost through its weaker direction. Since $\min(M_A, M_B) \le \frac{1}{2}(M_A + M_B) = \frac{1}{2}\sigma\|z_A - z_B\|^2$, with equality iff the potential and the rotation cancel on the pair, the symmetric pairs of (ii) are the most robust. For a fixed symmetric part J_s, key biases, key norms and rotations can only make a pair merge earlier, to leading order. Beyond leading order the critical value of the weaker margin depends on the partner margin. In units of $\kappa(t_0)^2$ it lies between 0.92 and 1.07 for partner margins up to 10, at $t_0 = 11$ and $t_0 = 10^3$ and for $\omega \in \{0, 1\}$ (Figure 3a). At every computed point the two critical margins sum to more than $2\kappa(t_0)^2$. So at a fixed gap the symmetric split is also the most robust one at finite t_0, as far as we computed.

What does not carry over. Two properties of the Gaussian dynamics fail for learned scores.

Proposition 6 (oscillation)**.** *Let $f(x, y) = x^\top J y$ and $\rho_A = \rho_B = \rho \ge 1$, and let the blocks start at $z_A = -z_B$ with $\sigma = e^\top J_s e < 0$. Then $z_A(t) = -z_B(t)$ for all t, and $D_t = (|\sigma|/2)^{1/2}\|z_A(t) - z_B(t)\|$ follows the Gaussian map F_t with cut ratios $(1/\rho, 1/\rho)$. If $D_0 \ge \kappa(t_0; 1/\rho, 1/\rho)$, the blocks exchange sides at every layer and $D_t \to D_\infty \ge 1$, so the features never converge, although the attention between the blocks is not summable. Otherwise the blocks merge.*

A score whose symmetric part is negative along a direction makes nodes prefer neighbours that differ along it, as a heterophilous attention head would. Under cooling such a head turns a two-block structure into a period-two oscillation, and both convergence and part (ii) of Proposition 3 fail. GAT [35] fails in a different way. Its score $\mathrm{LeakyReLU}(u^\top x + v^\top y)$ gives $|m_{ij} + m_{ji}| \le (1 - \beta)\min(|u^\top(z_i - z_j)|, |v^\top(z_i - z_j)|)$, where β is the negative slope, and $m_{ij} = -m_{ji}$ when the four arguments of the edge share a sign (Remark 6). In that regime one block of a pair has a nonpositive margin, and the gap shrinks by a fixed factor at every layer. This is the dynamical side of the static attention of GAT [2].

General graphs. The separation half of Theorem 5 holds for every score (Proposition 7). If the directed margins between adjacent blocks are at least δ^2 on the η-neighbourhoods of their hulls, and $e_{\max} R\,(t_0 - 1)^{1-\delta^2}/(\delta^2 - 1) \le \eta$ with R the largest diameter of the union of two such neighbourhoods, then every block stays in its neighbourhood and no two adjacent blocks merge. The in-block half of Theorem 5 and Proposition 3 rest on the symmetry of the weights, through the cut balance of Touri and Nedić [32]. Whether the features converge and the limiting partition consists of the components of zero-length edges when $J_s \succ 0$ is open. For prediction, we use the margin to the best neighbour, $m^+_{ij} = \max_{k \in N[i]} f(z_i, z_k) - f(z_i, z_j) \ge m_{ij}$, which sets the weight up to a bounded factor: $s_t^{-m^+_{ij}}/N_{\max} \le \alpha_{ij} \le s_t^{-m^+_{ij}}$. The *directed threshold-graph rule* returns the components of the edges with $\min(m^+_{ij}, m^+_{ji}) < \kappa(t_0 + t_1)^2$. For Gaussian attention it is the rule of Section 8. On two blocks it is the leading-order rule of Theorem 6.

Contextual SBMs (E9). We test the directed rule on the SBMs of E4 (four blocks, $p_{\text{out}} = 0.01$, $\sigma \in \{0.1, 0.2, 0.3\}$) under dot-product attention $f(x, y) = x^\top J y$, with tied ($J = 2I$) or untied queries and keys (Section F). With $J_s = 2I$ the symmetric part of every margin is the squared distance of E4, so a rule that sees only $\frac{1}{2}(m_{ij} + m_{ji})$ is the Gaussian rule. To create a key potential we shift all features by ν towards block 1. At the block means the margin of every other block towards block 1 then drops from 9 to $9 - 3\sqrt{2}\,\nu$, and the reverse margins grow by the same amount. The other blocks are absorbed into block 1 in 29 of 30 runs at $\nu = 3$ and in part of the runs at $\nu = 2$. The symmetric rule cannot see this: applied to the initial features, it predicts at least four clusters at every shift (Figure 3b). Applied after $t_1 = 20$ layers, the directed rule recovers the final partition with mean ARI

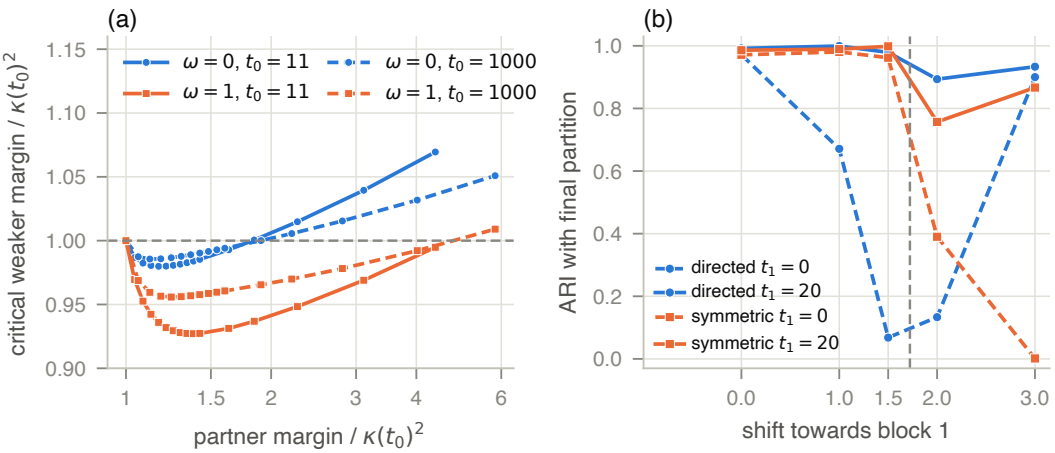


Figure 3: **Learned attention** (E9). (a) Critical value of the weaker margin of a two-block pair against the partner margin, both in units of $\kappa(t_0)^2$, for dot-product ($\omega = 0$) and Gaussian-type ($\omega = 1$) scores ($\rho_A = \rho_B = 1$; float bisection with the merge test of Theorem 6(i) and certificate C1′). The diagonal point is exact by Theorem 6(ii). (b) Contextual SBMs under dot-product attention, with all features shifted towards block 1 (30 runs per shift). ARI between the final partition and the directed rule, or the rule that uses only the symmetric part $\frac{1}{2}(m_{ij} + m_{ji})$, applied after t_1 layers. Dashed vertical line: shift at which the margin between block means equals $\kappa(t_0)^2$.

0.91 at $\nu \geq 2$, against 0.81 for the symmetric rule, and both reach 0.99 at $\nu \leq 1.5$. In 7 of the 10 runs where its ARI is below 0.9, the directed rule predicts fewer clusters than form.

At $t_1 = 0$ the directed rule over-merges. Its errors are one-sided: in 148 of the 150 runs no edge that it predicts to persist ends merged. The symmetric rule wrongly predicts persistence for some edges in 83 runs. The cause is that a learned margin is first order in the feature noise, through $\mathcal{V}(z_i) - \mathcal{V}(z_j)$, while the squared distance is second order. Before the blocks contract, a few cross edges of a well-separated pair fall below the threshold, and one such edge joins two predicted components. Five layers remove most of these errors: the mean ARI over all runs is 0.55, 0.93 and 0.96 at $t_1 = 0, 5, 20$. Blocks also contract towards their high-potential members before they interact, so the block-mean margin is only a rough guide: at $\nu = 2$ it is 0.51, below $\kappa^2 \approx 1.69$, yet most runs end with two to five clusters. No run oscillates, as expected for $J_s \succ 0$: the largest change in the last layer is $4 \cdot 10^{-4}$. Section 9 tests the rule on trained dot-product networks.

8 Predicting clusters

The theory suggests one prediction rule at three levels of generality. For chains, $\mathcal{R}_{t_1}$ thresholds every gap at $\kappa(t_0 + t_1)$ after t_1 layers (Theorem 2). For general graphs, the *threshold-graph rule* runs t_1 layers and returns the connected components of the edges with $\|z_i(t_1) - z_j(t_1)\| < \kappa(t_0 + t_1)$, using the two-layer κ. By Proposition 3 the true limiting partition consists of the components of zero-length edges. We test the rules against long runs (Section F).

Five-layer chains (E3). We draw 4000 random gap vectors of $G_{5,m}$ with gaps uniform in $[0.2, 2.6]$ at $t_0 = 17$ ($\kappa \in [1.4908, 1.4916]$) and certify the limiting partition of 3972 of them. The isolated rule classifies 91.3% of the 15,888 gaps and 68.0% of the partitions correctly. It never predicts persistence wrongly, as Theorem 2(b) requires. All 1387 errors are rescues, the smallest at $u_0 = 0.79$, below $\sqrt{c}$ (Figure 4a). The early-depth rule removes most of

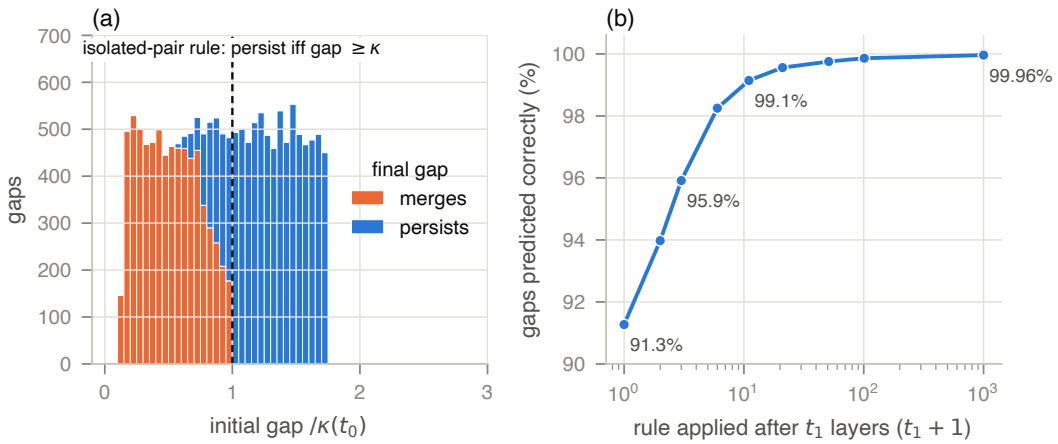


Figure 4: **Five-layer chains** ($t_0 = 17$; 3972 certified random initial conditions; 15,888 gaps). (a) Initial gaps, stacked by final fate. No gap at or above $\kappa(t_0)$ merges (Theorem 2). Every error of the isolated rule is a rescued gap below κ, down to $0.53\,\kappa$. (b) Accuracy of the early-depth rule $\mathcal{R}_{t_1}$ against t_1. By Theorem 2(c) it can only improve with t_1.

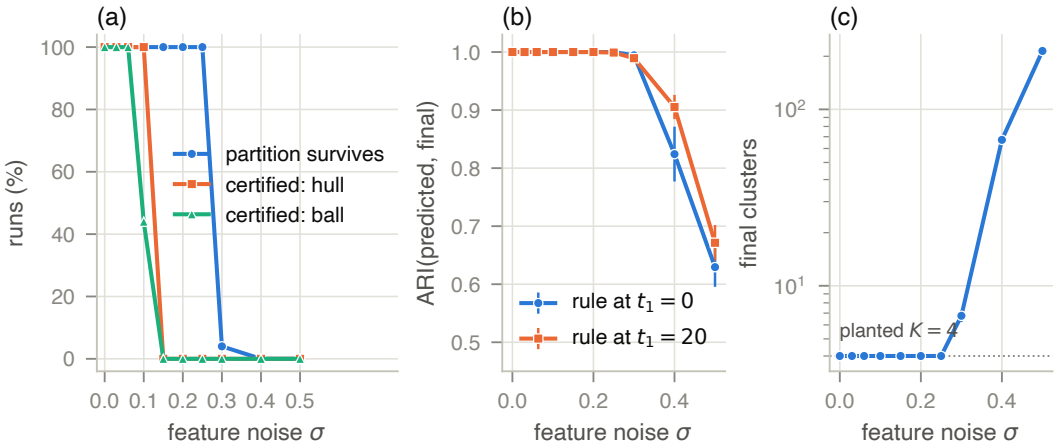


Figure 5: **Noisy contextual SBMs** ($K = 4$ blocks of 150 nodes, $p = 8$, block means $3\sqrt{c}$ apart, $t_0 = 602$, 4000 layers; 25 runs per σ). (a) Runs in which the planted partition survives, and runs certified by Theorem 5 (hull form) and by Corollary 2(a) (ball form). (b) ARI between the threshold-graph prediction and the final partition, with the rule applied at $t_1 = 0$ or after $t_1 = 20$ layers (mean ± 95% CI). (c) Number of final clusters (mean ± 95% CI).

them: it is correct for 94.0%, 98.3%, 99.1% and 99.96% of the gaps at $t_1 = 1, 5, 10$ and 1000 (Figure 4b).

Noisy community graphs (E4). We use $K = 4$ SBM blocks of 150 nodes ($p_{\text{in}} = 0.2$, $p_{\text{out}} \in \{0, 0.005, 0.01, 0.02, 0.05\}$), features in $\mathbb{R}^8$ with block means at mutual distance $3\sqrt{c}$ plus isotropic noise σ, and $t_0 = |V| + 2 = 602$, where $\kappa \approx 1.301$. We run 4000 layers and read off the limiting partition (Section F). The planted partition survives in every run with $\sigma \leq 0.25$. Beyond that it fragments, into ≈ 7 clusters at $\sigma = 0.3$, 67 at 0.4 and 214 at 0.5 (Figure 5c). The threshold-graph rule at $t_1 = 0$ recovers the final partition with ARI 1.00 up to $\sigma = 0.25$ and 0.99 at $\sigma = 0.3$, which includes predicting *which* blocks fragment and how. At $\sigma = 0.4$ and 0.5 the ARI drops to 0.82 and 0.63; applying the rule after $t_1 = 20$

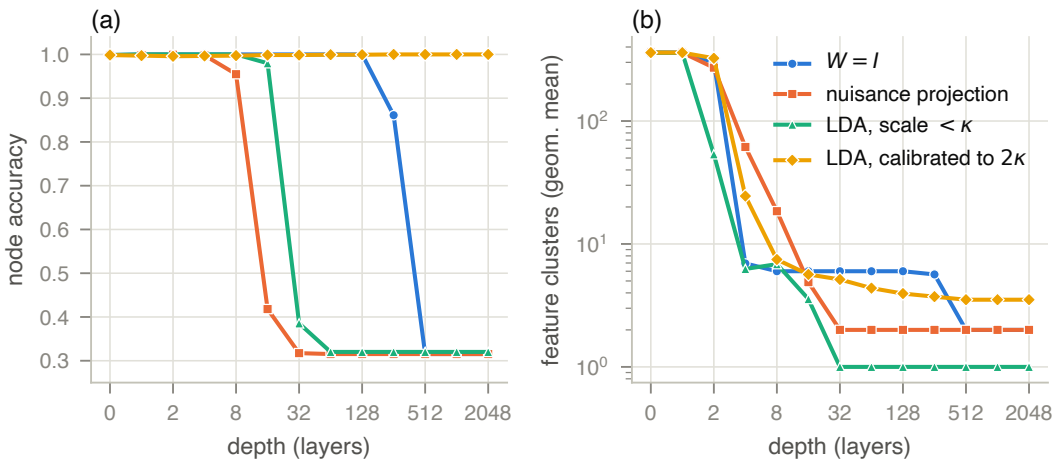


Figure 6: **A learned metric decides which distinctions survive** (5 seeds). Attention uses $z = Wh$, values are h. (a) Linear-probe accuracy against depth. (b) Number of final feature clusters (geometric mean over seeds). Only the metric that places the class distinctions above the threshold keeps the classes to depth 2048.

layers raises it to 0.90 and 0.67 (Figure 5b). Theorem 5 certifies survival for $\sigma \leq 0.1$. For $\sigma \geq 0.15$ every run has a block of diameter above $\sqrt{c}$, so by Proposition 4 no diameter-based theorem can certify these runs. Their survival up to $\sigma = 0.25$ comes from dense blocks and concentration of high-dimensional noise, which our theory does not capture.

A learned metric decides what survives (E5). By Remark 1, a metric W acts only through $\|W(h_i - h_j)\|$. We build three homophilous classes (3×120 nodes) whose means are $1.0\sqrt{c}$ apart in two signal dimensions, below the threshold $\kappa \approx 1.318$, plus a binary nuisance factor independent of the class at separation $3\sqrt{c}$ and four noise dimensions. With $W = I$, a linear probe on h is perfect up to depth 128 and at chance from depth 512 on: the class distinctions merge and only the nuisance split survives (Figure 6). Projecting onto the nuisance factor, or scaling an LDA metric so that class means are only 0.4κ apart, loses the classes by depth 32. Scaling the same LDA metric so that the closest class means are 2κ apart keeps accuracy 1.0 to depth 2048 and ends with 3–5 clusters. The threshold thus gives a concrete design rule: a learned metric must place the distinctions that should survive above $\sqrt{c}\,\kappa(t_0)$, and the distinctions that should be smoothed away below $\sqrt{c}$.

Real graphs (E8). We run the untrained dynamics for 4000 layers on Roman-empire (22,662 nodes, 18 classes, edge homophily 0.05, mean degree 2.9) and Amazon-ratings (24,492 nodes, 5 classes, homophily 0.38, mean degree 7.6) [27], with $t_0 = |V| + 2$, and on 300 test molecules of Peptides-func [8] (150 atoms on average, $t_0 = 152$). The values h are the first 64 principal components of the standardised node features (of the one-hot atom features for Peptides). Attention uses $z = Wh$ with two kinds of metric. The first takes the leading 16 principal components (8 for Peptides), scaled so that the median edge has length $s\,\kappa(t_0)$ with $s \in \{0.5, 1, 2\}$. The second is an LDA projection fitted on the training nodes, scaled so that the closest pair of class means is 2κ or 0.4κ apart, as in E5. Table 1 compares the threshold-graph rule with the partition after 4000 layers.

The rule recovers the final partition with ARI 0.79–1.00 on Roman-empire and 0.89–1.00 on the molecules. On Amazon-ratings, the densest graph, it gets 0.47–0.94. The errors are mostly of one kind, as on chains. Of the edges the rule predicts to persist, at most 1.6% end merged on Roman-empire and at most 5% on Amazon-ratings. Of the edges it predicts to merge, up to 40% end apart. On chains the first share is exactly zero (Theorem 2). On graphs it need not be: with vector features the centre of a cluster can be closer to a node

graph	metric	clusters	predicted	ARI, $t_1 = 0$ / 20		false persist / merge (%)	
Roman-empire	PCA, scale 0.5	5,758	5,682	0.985	0.994	0.2	0.7
	PCA, scale 1	10,150	8,985	0.874	0.882	1.6	13.9
	PCA, scale 2	17,206	16,651	0.786	0.803	1.4	16.5
	LDA, margin 2κ	22,591	22,591	1.000	1.000	0.0	0.0
	LDA, margin 0.4κ	22,282	22,036	0.713	0.712	0.0	40.5
Amazon-ratings	PCA, scale 0.5	718	427	0.678	0.733	5.0	3.0
	PCA, scale 1	6,173	4,927	0.534	0.593	3.3	19.5
	PCA, scale 2	18,268	17,161	0.465	0.454	0.6	28.5
	LDA, margin 2κ	24,319	24,300	0.944	0.956	0.0	11.8
	LDA, margin 0.4κ	21,652	20,805	0.647	0.670	0.1	39.3
Peptides-func	PCA, scale 0.5	1.1	1.1	1.000	1.000	–	–
	PCA, scale 1	92.8	76.2	0.893	0.790	–	–
	PCA, scale 2	129.7	129.1	0.986	0.986	–	–

Table 1: **Predicting clusters on real graphs** (E8, untrained dynamics, 4000 layers). *Clusters*: final partition; *predicted*: threshold-graph rule at $t_1 = 0$. *ARI*: between the rule applied after t_1 layers and the final partition. *False persist*: share of the edges predicted to persist (endpoints in different predicted clusters) whose endpoints end in one cluster. *False merge*: share of the edges predicted to merge whose endpoints end apart. Peptides-func: means over 300 test molecules.

than any member of the cluster is. Applying the rule after $t_1 = 20$ layers raises the ARI by up to 0.06 on the two node graphs. On the molecules at scale $s = 1$ it lowers the ARI from 0.89 to 0.79.

The design rule carries over. With the LDA metric at margin 2κ, a linear probe on Roman-empire keeps its depth-0 accuracy of 0.588 through 4000 layers. With the PCA metric the probe falls to 0.29 at scale 0.5 and to 0.53 at scale 2 (majority class: 0.14). On Amazon-ratings the linear probe is at the majority-class level (0.37) at every depth, so there is nothing to lose. One caveat concerns the reference partition itself. Near the threshold the dynamics slows down (Theorem 1(iv)). The ARI between the partitions at layers 2000 and 4000 is 0.92–1.00 on Roman-empire but only 0.61–0.99 on Amazon-ratings, so part of the measured error on Amazon-ratings is in the partition read off at layer 4000.

9 Deep networks in training

The theory describes the untrained dynamics (1). We now train it. Our Gaussian networks map the input to $z = Wx$ with a linear layer, apply K parameter-free layers of (1), and classify the final states with a linear readout. By Remark 1, W is a learned metric. The scale c of the cooled temperature $c/\log(l + 11)$, or a constant temperature T, is learned too. It is initialised at m times the median squared edge length, with $m \in \{1, 1/30\}$ chosen on validation data. We compare with an MLP, GCN [19], residual GCN, GCN with PairNorm [41], GCNII [6], APPNP [20], SGC [38], GAT [35], residual GAT and linear GRAND [5]. Each (task, model, depth, seed) picks the best of 8 configurations on validation accuracy: learning rate, dropout and one model-specific hyperparameter (Section F).

Synthetic tasks (E6b, E7). We use three contextual SBMs [7] with 3 classes of 150 nodes: homophilous with clean features, homophilous with noisy features, and heterophilous. We also use a long-range task: three rings of 200 nodes with random cross-class edges, where only 2% of the nodes carry class information and a typical node is 12 hops from the nearest informative one. Figure 7 shows test accuracy against depth $K \in \{2, \dots, 128\}$. GCN, SGC and GAT fall to chance by $K = 16$–32 on every task. Residual connections postpone the collapse to $K = 64$–128.

The Gaussian networks behave differently in a way the theory explains. By Proposition 3,

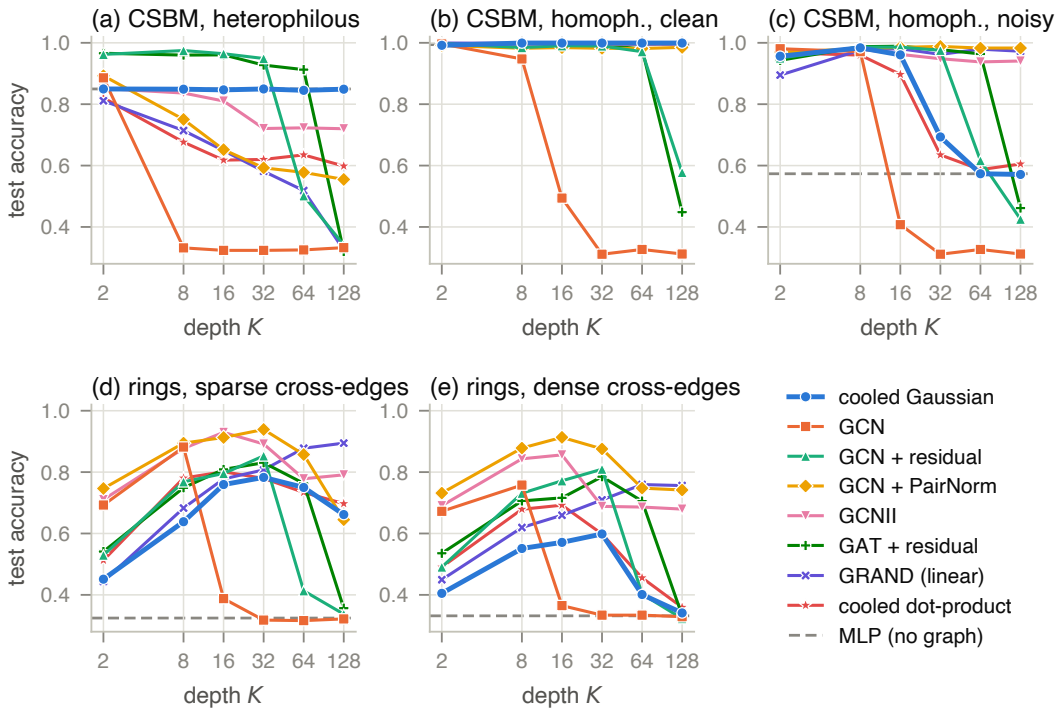


Figure 7: **Trained networks against depth** on the synthetic tasks (E6b, E7): mean test accuracy over 5 seeds, each run tuned over 8 configurations. (a–c) CSBMs with 3 classes; (d–e) ring task in which 2% of the nodes carry the class. Dashed: MLP. All 13 models with confidence intervals are in Tables 6 and 7; stars: the cooled dot-product network of Section 7. Gaussian attention with constant temperature is within 0.007 of the cooled version on the CSBMs and up to 0.10 better on the rings at depth $K \geq 32$.

a deep network of this kind ends in a partition of the nodes into clusters, and training chooses the partition through W and c. Two partitions are always available. One cluster per connected component is oversmoothing. The partition into singletons, the frozen phase, reproduces the MLP. Which phase a run reaches is set mostly by the initial temperature.

- In the heterophilous CSBM, validation picks the cold start $m = 1/30$ in all 30 runs (6 depths, 5 seeds) of each Gaussian model. The network then matches the MLP seed by seed, within 0.015 in test accuracy. At $K = 128$ the Gaussian networks are the only models at the MLP level of 0.85. Every baseline is at least 0.12 below it.

- In the clean homophilous CSBM, the default start reaches the class partition and keeps it. The cooled network has accuracy 0.999 at every depth from 8 to 128, the best or within 0.001 of the best.

- In the noisy homophilous CSBM the class partition is not reachable at depth. The default start gives 0.984 at $K = 8$, within 0.003 of the best model. From $K = 64$ on it collapses to chance (0.33, effective rank 1.5–3), and the cold start freezes at the MLP level (0.57). At $K = 32$ two of five seeds still reach the class partition.

The noisy regime is the case in which the design rule of Section 8 cannot be met. Class means are 1.5 apart and the noise has standard deviation 1 in every coordinate. Along any direction, a large share of the within-class edges is about as long as the distance between class means, so no metric separates the two scales by the factor κ that the rule needs. A shallow network stops before the limit and can still smooth partially. A deep one cannot.

On the ring task the Gaussian networks improve with depth up to $K = 32$, but stay well below PairNorm and GCNII. The cooled network peaks at 0.60 (dense cross-edges) and 0.78 (sparse), against 0.91 and 0.94 for PairNorm. Similarity-based attention has nothing to work with at nodes whose features are pure noise. Here the constant temperature is better at depth, by up to 0.10, consistent with cooling sharpening the attention and so shortening the range of propagation.

Cooled dot-product attention. The same network with the learned score $\langle W_Q z_i, W_K z_j \rangle$ of Section 7 ($J = W_Q^\top W_K$, shared by all layers) behaves like the Gaussian one where a coarse partition is wanted. It reaches 0.995–0.998 at every depth on the clean CSBM, and on the rings it peaks higher, at 0.80 (sparse) and 0.69 (dense) for $K = 16$, but still below PairNorm. The difference is the frozen phase. A dot-product score need not be largest at $j = i$, so some margins can be negative, and then no temperature makes every node keep its own state (Proposition 5). On the heterophilous CSBM, validation picks the cold start in 28 of 30 runs and the network still falls from 0.82 at $K = 2$ to 0.60–0.68 for $K \geq 8$, against 0.85 for the MLP and the Gaussian networks. On the noisy CSBM it holds 0.96 to $K = 8$ and drops to about 0.6 from $K = 32$ on.

Do trained networks reach the predicted partition? For each of the 150 reported dot-product runs we rerun the network's own dynamics, with its learned J and c, for 4000 layers from its input states (Section F); at layer K this reproduces the trained network's test accuracy to $3 \cdot 10^{-8}$. The two starts then behave differently.

- The 46 cold-start runs are close to their limit at depth K. Run to 4000 layers they lose 0.004 test accuracy on average (at most 0.048), and every one settles into a partition with a few to several hundred clusters. The directed rule applied after $t_1 = 20$ layers recovers it with mean ARI 0.97, below 0.9 in 2 runs, and the symmetric rule gets 0.96. Applied at $t_1 = 0$ both fail (ARI 0.13 and 0.00). In trained networks the rule thus needs a few layers before it applies, as in E9, and five are not enough (ARI 0.66).
- The 104 default-start runs rely on the transient. Run to 4000 layers they lose 0.29 test accuracy on average and up to 0.70. In 47 of them, all on the rings or the clean CSBM, the partition is still changing at layer 4000. In the 57 others the final partition has one to a few clusters, and the directed rule at $t_1 = 20$ gets mean ARI 0.88 (symmetric rule 0.94). Its errors are again one-sided: in 56 of the 57 runs no edge that it predicts to persist ends merged, and most misses are runs in which it predicts fewer clusters than form, often one where two form.

Training makes J_s indefinite in 37 runs, 31 of them on the rings, with $\lambda_{\min}(J_s)$ down to -1.8. None of them oscillates: the largest change in the last layer is $4 \cdot 10^{-5}$. Proposition 6 is for a specific configuration, and trained networks do not reach one. A trained network at depth K is thus either near its limiting partition, which the directed rule predicts after a few layers, or in a transient whose accuracy the limit does not keep.

Real graphs (E8). We train the Gaussian networks and the ten baselines on Roman-empire and Amazon-ratings with $K \in \{2, 4, 8, 16, 32\}$ and width 64, on three of the ten official splits, with the learning rate tuned per model and depth on split 0 (Table 8). Roman-empire repeats the heterophilous CSBM. The cooled network freezes, ending with 21,500–22,200 clusters on 22,662 nodes, and holds the MLP accuracy (0.642–0.643 against 0.644) at every depth, while GCN, SGC and GAT fall to 0.13–0.19 by $K = 32$. Residual GCN and residual GAT are much better at moderate depth (0.79–0.80 at $K = 4$–8) but drop to 0.60–0.63 at $K = 32$. On Amazon-ratings the cooled network (0.415–0.419 for $K \leq 16$) beats the MLP (0.386) but not PairNorm (0.470).

Run to 4000 layers, the trained cooled Gaussian networks fall into the same two groups as the dot-product networks. On Roman-empire their test accuracy changes by at most 0.006,

and the threshold-graph rule applied after K layers recovers the final partition with ARI 0.79–0.99. On Amazon-ratings every network falls to the majority-class level (0.37–0.38), and its partition is still changing at layer 4000 (ARI between layers 2000 and 4000 as low as 0.12): these networks use the transient. On Peptides-func, a long-range molecular benchmark, propagation helps (Table 9, one seed, $K \in \{2, 8, 32\}$). Residual GAT reaches average precision 0.421 at $K = 8$ against 0.253 for the MLP, and plain GCN falls from 0.349 at $K = 2$ to 0.169 at $K = 32$, below the MLP. Both Gaussian networks stay at the MLP level (0.246–0.254) at every depth, cooled or not: on these graphs they keep the MLP's accuracy but do not use the graph.

Summary. The cooled Gaussian network is an analysable model, not a competitive architecture. Because the frozen phase is always reachable, its mean accuracy is never more than 0.005 below the MLP's at any depth on the seven node tasks, and its average precision on Peptides-func never more than 0.007 below; residual and normalised baselines beat it wherever propagation helps. What the theory adds is prediction: which distinctions survive, which metric keeps them, and whether the accuracy of a trained network at depth K is a property of its limit or of a transient.

10 Discussion and limitations

What is proved and what is not. Theorems 1 to 3 and 5 and Propositions 2 to 4 and Corollary 1 are analytic. So are Theorem 6 and Propositions 5 to 7. Theorem 4 is computer-assisted. The survival of noisy blocks with diameter above $\sqrt{c}$ (E4) is an observation without proof, and so is the dependence of the critical weaker margin on the partner margin in Figure 3(a), which is a float computation. The scaling limit in Theorem 3(c) holds beyond an explicit but astronomically large t_0 near the corner $x = 2/3$. The quantitative chain theory is for scalar, layer-constant features; Proposition 3 and Theorem 5 cover vector features on general graphs, but only through sufficient conditions.

Model scope. (1) has no value matrix, nonlinearity, residual connection or multiple heads. Remark 1 covers a fixed learned metric and Section 7 a fixed quadratic query–key score. A layer-dependent score changes the margins layer by layer, and our rules then apply only between changes. For learned scores the exact theory is for two blocks; on general graphs we have only the separation result of Proposition 7, and whether Proposition 3 extends to scores with $J_s \succ 0$ is open. Logarithmic cooling is a modelling choice that makes the threshold finite and nontrivial. We do not claim that it is the best schedule for training.

Using the theory. The early-depth rule is a cheap diagnostic for distance-based and query–key attention: run a few layers, threshold the directed margins at $\kappa(t_0 + t_1)^2$, and read off the partition the network converges to. Comparing it with the labels shows which distinctions depth will erase, and the design rule says how to scale a metric so that it keeps the right ones. In trained networks the same comparison tells whether accuracy at the trained depth will survive further depth. Extending the theory to layer-dependent scores, value maps and residual connections is the main open step.

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A Proofs for Sections 3 and 4

Proof of Proposition 1. Each $h_i(t+1)$ is a convex combination of $\{h_j(t)\}$, so the feature diameter R_t is nonincreasing and $\alpha_{ij}(t) \geq e^{-\|W\|^2 R_0^2/T}/N_{\max} =: \gamma > 0$ on every edge, including self-loops. A product of stochastic matrices whose positive entries are bounded below by γ on a fixed connected graph with self-loops is a consensus product, at an exponential rate [24]. Indeed, every product of $\operatorname{diam}(G)$ consecutive matrices has a column bounded below by $\gamma^{\operatorname{diam}(G)}$, so its Dobrushin coefficient is at most $1-\gamma^{\operatorname{diam}(G)}$. □

Proof of Lemma 1. Let z_A, z_B be the block values and $a = s_t^{-d^2}$. A node of A sees k_A nodes at distance 0 and e_A nodes at distance d, so $z'_A = (k_A z_A + e_A a\, z_B)/(k_A + e_A a) = z_A + \frac{\rho_A a}{1+\rho_A a}(z_B - z_A)$, and symmetrically for B. All nodes of a block receive the same update, so the features stay block-constant, and

$$z'_B - z'_A = (z_B - z_A)\Big(1 - \frac{\rho_A a}{1+\rho_A a} - \frac{\rho_B a}{1+\rho_B a}\Big) = (z_B - z_A)\,\frac{1 - Pa^2}{(1+\rho_A a)(1+\rho_B a)}. \qquad \square$$

Properties of F_t. For $a \in [0,1)$ we have $0 < \psi(a) \leq \frac{1-Pa^2}{1+\lambda a+Pa^2} \leq \frac{1}{1+\lambda a} \leq 1$, because $P \leq 1$. The function ψ is decreasing in a and nonincreasing in ρ_A and ρ_B. Since $a = s_t^{-d^2}$ decreases in d, $F'_t(d) = \psi(a) + d\,\psi'(a)\,\partial_d a > 0$. So F_t is a continuous increasing bijection of $(0,\infty)$ with $F_t(d) \leq d$. For fixed d and t, $F_t(d)$ is nondecreasing in t_0 and nonincreasing in ρ_A, ρ_B. Write $\Phi_k = F_{k-1} \circ \cdots \circ F_0$, so that $d_k = \Phi_k(d_0)$, and note that the trajectory from d_0 at offset t_0 continues after k steps as the trajectory from $\Phi_k(d_0)$ at offset $t_0 + k$.

Certificate C1 (Theorem 1(vi), second half). Suppose $d_u \geq \delta$ for all $u < t$. Then $a_u \leq s_u^{-\delta^2} \leq q$, and

$$-\log\psi(a) = \log(1+\rho_A a) + \log(1+\rho_B a) - \log(1-Pa^2) \leq \lambda a + \frac{Pa^2}{1-Pa^2} \leq \Lambda_\delta\, a.$$

With $\sum_{u\geq 0}(t_0+u)^{-\delta^2} \leq \int_{t_0-1}^{\infty} x^{-\delta^2}dx = \frac{(t_0-1)^{1-\delta^2}}{\delta^2-1}$ we get $\log d_t \geq \log d_0 - \Lambda_\delta \frac{(t_0-1)^{1-\delta^2}}{\delta^2-1} \geq \log\delta$, and induction on t closes the argument.

Proof of Theorem 1. (i) The sequence d_t is positive and nonincreasing, so it converges. If $d_T \leq 1$, then $d_t \leq 1$ and $a_t \geq s_t^{-1}$ for all $t \geq T$. Hence $d_{t+1} \leq d_t/(1+\lambda a_t) \leq d_t \exp(-\frac{\lambda}{1+\lambda} s_t^{-1})$, and $\sum_t s_t^{-1} = \infty$ gives $d_t \to 0$. Otherwise $d_t > 1$ for all t and $d_\infty \geq 1$.

(ii) By (i), the merge set is $M(t_0) = \bigcup_T \{d_0 : \Phi_T(d_0) \leq 1\}$. If $\Phi_T(d_0) = 1$ then $\Phi_{T+1}(d_0) < 1$, so $M(t_0) = \bigcup_T \Phi_T^{-1}((0,1)) = \bigcup_T (0, \Phi_T^{-1}(1))$, an open interval $(0,\kappa)$. Since $F_0(1) < 1$, $1 \in M(t_0)$ and $\kappa > 1$. Finiteness follows from (vi). Since $\kappa \notin M(t_0)$, the threshold gap itself persists.

(iii) $d_\infty(d_0) = \inf_T \Phi_T(d_0)$ is nondecreasing and, as an infimum of continuous functions, upper semicontinuous. A nondecreasing upper semicontinuous function is right-continuous. Since κ persists, $d_\infty(\kappa) \geq 1$. Suppose $d_\infty(\kappa) = 1 + 2\eta$ with $\eta > 0$, and let $\delta = 1 + \eta/2$. Because $(t_0 + T - 1)^{1-\delta^2} \to 0$, there is T such that (5) holds at offset $t_0 + T$ for every initial value $\geq 1 + \eta$. Since $\Phi_T(\kappa) \geq 1 + 2\eta$ and Φ_T is continuous, some $d_0 < \kappa$ has $\Phi_T(d_0) \geq 1 + \eta$. By C1 at offset $t_0 + T$, the trajectory from d_0 stays above $\delta > 1$, so d_0 persists. This contradicts $d_0 \in M(t_0) = (0,\kappa)$, so $d_\infty(\kappa) = 1$.

(iv) $M(t_0 + k) = \Phi_k(M(t_0)) = (0, \Phi_k(\kappa(t_0)))$, because Φ_k is an increasing bijection. For monotonicity, let $t_0 \leq t'_0$ (or $\rho \geq \rho'$ componentwise) and compare the trajectories d_t, d'_t from the same d_0: $d'_{t+1} = F'_t(d'_t) \geq F'_t(d_t) \geq F_t(d_t) = d_{t+1}$. Hence $M(t'_0) \subseteq M(t_0)$ (respectively $M(\rho') \subseteq M(\rho)$).

(vi), lower bound. Let $d_0 = 1+\varepsilon$ persist, for instance $\varepsilon = \kappa - 1$. Then $1 \le d_\infty \le d_t \le d_0$, so $s_t^{-1} \ge a_t \ge s_t^{-(1+\varepsilon)^2}$. Using $\log(1+y) \ge y/(1+y)$ and $a_t \le 1/t_0$,

$$\begin{aligned}\log(1+\varepsilon) \;&\ge\; \log\frac{d_0}{d_\infty} = \sum_t -\log\psi(a_t) \;\ge\; \sum_t \big[\log(1+\rho_A a_t) + \log(1+\rho_B a_t)\big] \\ &\ge\; \lambda_- \sum_{t\ge 0} s_t^{-(1+\varepsilon)^2} \;\ge\; \lambda_- \,\frac{t_0^{-y}}{y},\end{aligned}$$

with $y = 2\varepsilon + \varepsilon^2$. Since $\log(1+\varepsilon) \le \varepsilon$, we get $\varepsilon(2\varepsilon+\varepsilon^2) \ge \lambda_- t_0^{-(2\varepsilon+\varepsilon^2)}$. The left side increases and the right side decreases in ε, so $\varepsilon \ge \varepsilon_-$.

Part (v) is proved below. □

A.1 Proof of Theorem 1(v): barriers

Write $\tau_t = \log s_t$ and $\Delta_t = \tau_{t+1} - \tau_t = \log(1+e^{-\tau_t}) \in \big[e^{-\tau_t}(1-\frac{1}{2}e^{-\tau_t}),\, e^{-\tau_t}\big]$. Let $c(\tau) = c_0 + (\beta \log\log\tau + k)/\log\tau$ with constants c_0, β, k (the proof of (v) uses $\beta = k = 0$, that of Proposition 2 uses $\beta = \frac{1}{4}$), and for $\tau > e$ let

$$N_c(\tau) = \log\tau - \tfrac{1}{2}\log\log\tau + c(\tau), \qquad X_c(\tau) = N_c(\tau)/\tau, \qquad \text{so} \qquad e^{-2\tau X_c(\tau)} = e^{-2c(\tau)}\,\frac{\log\tau}{\tau^2}. \tag{$*$}$$

Lemma 3 (barrier functions)**.** *There is τ_1 such that on $[\tau_1,\infty)$: $0 < X_c \le 1$, $X_c' < 0 < X_c''$, $\tau X_c^2 \le 1$, and*

$$\frac{\tau^2|X_c'(\tau)|}{\log\tau} = 1 - r_c(\tau), \qquad r_c(\tau) = \frac{\frac{1}{2}\log\log\tau - c(\tau) + 1 - \frac{1}{2\log\tau} + \tau c'(\tau)}{\log\tau},$$

where $r_c(\tau) \to 0$ and $\tau c'(\tau) = O(\log\log\tau/(\log\tau)^2)$. Moreover $\tau X_c(\tau)^2 \to 0$ and $r_c(\tau+\Delta) = r_c(\tau) + O(e^{-\tau})$ for $0 \le \Delta \le e^{-\tau}$.

Proof. With $u = \log\tau$, $\tau c' = (\beta - \beta\log u - k)/u^2$. Then $N_c' = (1 - \frac{1}{2\log\tau})/\tau + c'$, so $\tau^2 X_c' = \tau N_c' - N_c = -(N_c - 1 + \frac{1}{2\log\tau} - \tau c')$, which gives the formula for r_c and $X_c' < 0$ for large τ. Further $\tau^3 X_c'' = 2N_c - 2\tau N_c' + \tau^2 N_c'' = 2\log\tau + O(\log\log\tau) > 0$, $\tau X_c^2 = N_c^2/\tau \to 0$, and $r_c'(\tau) = O(1/\tau)$. □

For $0 \le a \le e^{-\tau}$ and $\tau \ge 1$ we use two bounds on ψ. Since $\psi(a) \ge (1-Pa^2)(1-\lambda a - Pa^2) \ge 1 - \lambda a - 2Pa^2$ and $P \le 1$,

$$1 - \psi(a) \le a\,(\lambda + 2e^{-\tau}). \tag{8}$$

Since $1 - \psi(a) = \frac{\lambda a + 2Pa^2}{1+\lambda a + Pa^2} \ge \frac{\lambda a}{1+(\lambda+1)a}$,

$$1 - \psi(a) \ge \lambda a\,\big(1 - (\lambda+1)e^{-\tau}\big). \tag{9}$$

Proof of Theorem 1(v). Let $L = \log t_0$ and fix $\eta > 0$.

Upper bound. Let $X = X_c$ with $c \equiv \frac{1}{2}\log\lambda + \eta$. We claim that there is $\tau_2 \ge \tau_1$ with

$$F_t\big(1 + X(\tau_t)\big) \;\ge\; 1 + X(\tau_{t+1}) \qquad \text{whenever } \tau_t \ge \tau_2. \tag{10}$$

Write $\tau = \tau_t$, $\Delta = \Delta_t$ and $d = 1 + X(\tau)$. Then $a = s_t^{-d^2} = e^{-\tau}e^{-2\tau X}e^{-\tau X^2} \le e^{-\tau}\lambda^{-1}e^{-2\eta}\log\tau/\tau^2$ by $(*)$, and (8) gives

$$d - F_t(d) = d\,(1-\psi(a)) \;\le\; e^{-\tau}\,(1+X)\Big(1 + \frac{2e^{-\tau}}{\lambda}\Big)\,e^{-2\eta}\,\frac{\log\tau}{\tau^2}.$$

By convexity, $X(\tau)-X(\tau+\Delta)\ge\Delta\,|X'(\tau+\Delta)|\ge e^{-\tau}(1-\frac12 e^{-\tau})|X'(\tau+\Delta)|$, and by Lemma 3 $|X'(\tau+\Delta)|=(1+o(1))\log\tau/\tau^2$ as $\tau\to\infty$, uniformly in $\Delta\in[0,e^{-\tau}]$. Since $e^{-2\eta}<1$ and $(1+X)(1+2e^{-\tau}/\lambda)\to1$, we get $X(\tau)-X(\tau+\Delta)\ge d-F_t(d)$ for large τ, which is (10). Now let $L\ge\tau_2$ and $d_0\ge1+X(L)$. Since F_t is increasing, induction with (10) gives $d_{t+1}=F_t(d_t)\ge F_t(1+X(\tau_t))\ge1+X(\tau_{t+1})>1$ for all t. By (i) the gap persists. Hence $\kappa(t_0)\le1+X_{\frac12\log\lambda+\eta}(L)$ for all large t_0.

Lower bound. Let $Y=X_c$ with $c\equiv\frac12\log\lambda-\eta$. We claim that there is $\tau_3\ge\tau_1$ with

$$F_t\big(1+Y(\tau_t)\big)\;\le\;1+Y(\tau_{t+1})\qquad\text{whenever }\tau_t\ge\tau_3. \tag{11}$$

With $d=1+Y(\tau)$ we have $a=e^{-\tau}\lambda^{-1}e^{2\eta}e^{-\tau Y^2}\log\tau/\tau^2\le e^{-\tau}$ for large τ, and (9) gives

$$d-F_t(d)\;\ge\;e^{-\tau}\,(1+Y)\,e^{2\eta}\,e^{-\tau Y^2}\big(1-(\lambda+1)e^{-\tau}\big)\frac{\log\tau}{\tau^2},$$

while convexity gives $Y(\tau)-Y(\tau+\Delta)\le\Delta|Y'(\tau)|\le e^{-\tau}(1-r_c(\tau))\log\tau/\tau^2$. Since $e^{2\eta}>1$ and $\tau Y^2\to0$, this proves (11) for large τ. Now let $L\ge\tau_3$ and $d_0=1+Y(L)$. By induction, $d_t\le1+Y(\tau_t)$ for all t. Suppose the gap persists. By (i), $d_t>1$ for all t, so $s_t^{-(1+Y(\tau_t))^2}\le a_t\le e^{-\tau_t}$. By (9) and $(*)$,

$$d_t-d_{t+1}=d_t(1-\psi(a_t))\;\ge\;\lambda a_t\big(1-(\lambda+1)e^{-\tau_t}\big)\;\ge\;e^{2\eta}\big(1-\epsilon(\tau_t)\big)\,g(s_t),\qquad g(s)=\frac{\log\log s}{s\,(\log s)^2},$$

with $\epsilon(\tau)=1-e^{-\tau Y(\tau)^2}(1-(\lambda+1)e^{-\tau})\to0$. Choose $\tau_4\ge\tau_3$ such that $1-\epsilon(\tau)\ge e^{-\eta}$ and g is decreasing for $\log s\ge\tau_4$, and let $L\ge\tau_4$. Summing over t and substituting $\tau=\log s$,

$$\begin{aligned}d_0-1\;&\ge\;\sum_{t\ge0}(d_t-d_{t+1})\;\ge\;e^{\eta}\sum_{t\ge0}g(t_0+t)\;\ge\;e^{\eta}\int_{t_0}^{\infty}g(s)\,ds\\&=\;e^{\eta}\int_L^{\infty}\frac{\log\tau}{\tau^2}\,d\tau\;=\;e^{\eta}\,\frac{\log L+1}{L}.\end{aligned}$$

But $d_0-1=Y(L)=(\log L-\frac12\log\log L+\frac12\log\lambda-\eta)/L$, which is smaller than $e^{\eta}(\log L+1)/L$ for large L. This is a contradiction, so $d_0=1+Y(L)$ merges, and $\kappa(t_0)>1+X_{\frac12\log\lambda-\eta}(L)$ for all large t_0.

Since $\eta>0$ was arbitrary, $(\kappa-1)L-(\log L-\frac12\log\log L+\frac12\log\lambda)\to0$. In physical units $d_{\mathrm{crit}}^2=c\kappa^2=c+2c(\kappa-1)+c(\kappa-1)^2$, where $c/L=T_0$, $\log L=\log(c/T_0)$ and $c(\kappa-1)^2=T_0\,O((\log L)^2/L)=o(T_0)$. □

Proof of Corollary 1. Subtract the expansions of (v) for (ρ_A,ρ_B) and (ρ'_A,ρ'_B); in physical units multiply by $2c\kappa$, using $c/\log t_0=T_0$ and $\kappa\to1$. □

Remark 3 (where the third term comes from). In the continuum limit $dx/d\tau=-\lambda e^{-2\tau x}$ (dropping terms of relative order x and $e^{-\tau}$), write $x=(N_{\frac12\log\lambda}(\tau)+y/2)/\tau$. Then

$$\frac{\tau y'}{2}=\log\tau\,(1-e^{-y})-\tfrac12\log\log\tau+\tfrac12\log\lambda-1+\frac{y}{2}+\frac{1}{2\log\tau}.$$

The linearisation $\tau\,\delta y'\approx2\log\tau\,\delta y$ shows that the critical solution is repelling, so it stays near the zero of the right side, $y^*=(\log\log\tau-\log\lambda+2)/(2\log\tau)+O\big((\log\log\tau)^2/(\log\tau)^2\big)$. This is the third term of Proposition 2; the barrier proof below makes it rigorous for the discrete map.

Proof of Proposition 2. Fix $\eta>0$, let $\delta_\pm(\tau)=(\log\log\tau-\log\lambda+2\pm\eta)/(4\log\tau)$ and $c_\pm=\frac12\log\lambda+\delta_\pm$, and let $X=X_{c_+}$ and $Y=X_{c_-}$. Since $\delta_\pm=O(\log\log\tau/\log\tau)$, Lemma 3

gives

$$1 - r_{c_\pm}(\tau) = 1 - \frac{\frac{1}{2}\log\log\tau - \frac{1}{2}\log\lambda + 1}{\log\tau} + O\Big(\frac{\log\log\tau}{(\log\tau)^2}\Big),$$
$$e^{-2\delta_\pm(\tau)} = 1 - \frac{\frac{1}{2}\log\log\tau - \frac{1}{2}\log\lambda + 1 \pm \frac{\eta}{2}}{\log\tau} + O\Big(\frac{(\log\log\tau)^2}{(\log\tau)^2}\Big),$$

so that

$$\big(1 - r_{c_\pm}(\tau)\big) - e^{-2\delta_\pm(\tau)} = \pm\frac{\eta}{2\log\tau} + O\Big(\frac{(\log\log\tau)^2}{(\log\tau)^2}\Big). \tag{12}$$

We repeat the proof of (v), with $e^{\mp 2\eta}$ replaced by $e^{-2\delta_\pm(\tau)}$ in the bounds on a.

Upper bound. As there, (10) holds if

$$\big(1 - \tfrac{1}{2}e^{-\tau}\big)\big(1 - r_{c_+}(\tau+\Delta)\big)\frac{\log(\tau+\Delta)}{(\tau+\Delta)^2} \;\geq\; (1+X)\Big(1 + \frac{2e^{-\tau}}{\lambda}\Big)e^{-2\delta_+(\tau)}\frac{\log\tau}{\tau^2}.$$

By Lemma 3, every factor other than $1 - r_{c_+}(\tau)$ and $e^{-2\delta_+(\tau)}$ is $1 + O(\log\tau/\tau)$, so this follows from (12) for large τ. Hence $\kappa(t_0) \leq 1 + X(L)$ for large t_0.

Lower bound. As there, (11) holds if $(1+Y)\,e^{-2\delta_-(\tau)}e^{-\tau Y^2}(1 - (\lambda+1)e^{-\tau}) \geq 1 - r_{c_-}(\tau)$, which follows from (12) in the same way. Let $d_0 = 1 + Y(L)$, so that $d_t \leq 1 + Y(\tau_t)$ for all t by induction, and suppose that the gap persists. Then $d_t - d_{t+1} \geq e^{-2\delta_-(\tau_t)}(1 - \epsilon(\tau_t))\,g(s_t)$ with $\epsilon(\tau) \leq \tau Y(\tau)^2 + (\lambda+1)e^{-\tau} = O((\log\tau)^2/\tau)$. For large τ, δ_- is decreasing, so with $\bar\epsilon(L) = \sup_{\tau\geq L}\epsilon(\tau)$,

$$\begin{aligned} L\,(d_0 - 1) \;&\geq\; e^{-2\delta_-(L)}\big(1 - \bar\epsilon(L)\big)(\log L + 1) \;\geq\; \log L + 1 - 2\delta_-(L)\log L - O\Big(\frac{\log\log L}{\log L}\Big) \\ &=\; \log L - \tfrac{1}{2}\log\log L + \tfrac{1}{2}\log\lambda + \frac{\eta}{2} - o(1), \end{aligned}$$

using $e^{-x} \geq 1 - x$. But $L(d_0 - 1) = LY(L) = \log L - \frac{1}{2}\log\log L + \frac{1}{2}\log\lambda + \delta_-(L)$ and $\delta_-(L) \to 0$, a contradiction for large L. Hence $\kappa(t_0) > 1 + Y(L)$.

Since $\eta > 0$ was arbitrary, $(\kappa - 1)L = \log L - \frac{1}{2}\log\log L + \frac{1}{2}\log\lambda + \frac{\log\log L - \log\lambda + 2 + o(1)}{4\log L}$. For the topology shift, subtract the expansions for λ and λ': the third terms differ by $-\log(\lambda/\lambda')/(4\log L)$. □

B Proofs for chains

Derivation of (4). By symmetry, layer-constant features stay layer-constant. A node of layer l sees m nodes of layers $l-1$, l, $l+1$ with weights $w_{l-1}, 1, w_l$, so $x_l' = (w_{l-1}x_{l-1} + x_l + w_l x_{l+1})/D_l$. Then $x_l' - x_l = (w_l g_l - w_{l-1}g_{l-1})/D_l$, and $g_l' = x_{l+1}' - x_l'$ equals $g_l\big(1 - \frac{w_l}{D_l} - \frac{w_l}{D_{l+1}}\big) + \frac{w_{l+1}g_{l+1}}{D_{l+1}} + \frac{w_{l-1}g_{l-1}}{D_l}$. With $D_l = A + w_l$ and $D_{l+1} = B + w_l$, the bracket equals $(AB - w_l^2)/(D_l D_{l+1})$.

Proof of Lemma 2. All terms of (4) are nonnegative, since $AB \geq 1 \geq w_l^2$. It remains to show $(AB - w^2)(1+w) \geq (1-w)(A+w)(B+w)$ for $A, B \geq 1$ and $w \in [0,1]$; the difference equals $w\,[(A-1)B + A(B-1) + w(A+B-2)] \geq 0$. Dropping the push terms gives $g_l' \geq g_l\frac{1-w_l}{1+w_l} = F_t(g_l)$. □

Proof of Theorem 2. (a) By Lemma 2 and the monotonicity of F_t, induction gives $g_l(t_1 + k) \geq d_k$, where d is the isolated trajectory at offset $t_0 + t_1$ from $\kappa(t_0 + t_1)$. By Theorem 1(ii),(iii), $d_k > 1$ for all k. (b) is (a) with $t_1 = 0$. (c) One-sidedness is (a). If $g_l(t_1) \geq \kappa(t_0 + t_1)$, then by Lemma 2 and Theorem 1(iv), $g_l(t_1 + 1) \geq F_{t_1}(g_l(t_1)) \geq F_{t_1}(\kappa(t_0 + t_1)) = \kappa(t_0 + t_1 + 1)$, where F_{t_1} is taken at offset t_0. So the declared set is nondecreasing. A merging gap has $g_l(t_1) \to 0 < 1 < \kappa(t_0 + t_1)$ and is always declared merging. A persisting gap with $g_l(t) \to g_\infty > 1$ is eventually declared persisting, because $\kappa(t_0 + t_1) \to 1$ by Theorem 1(v). □

B.1 Three layers

Let $u = x_2 - x_1$, $v = x_3 - x_2$, $w_1 = s_t^{-u^2}$, $w_2 = s_t^{-v^2}$, $D_1 = 1 + w_1$, $D_2 = 1 + w_1 + w_2$, $D_3 = 1 + w_2$. Write $D := u + v$, $h := u + v/2$ and $\tilde{h} := v + u/2$. By Proposition 3, the limits u_∞, v_∞ exist and lie in $\{0\} \cup [1, \infty)$. Define

$$U_+(v_0; t_0) := \inf\{u : \text{ gap 1 persists from } (u', v_0)\ \forall u' \geq u\},$$
$$U_-(v_0; t_0) := \sup\{u : \text{ gap 1 merges from } (u', v_0)\ \forall u' \leq u\},$$

so that $U_- \leq U_+$ and $u^* \in [U_-, U_+]$.

Lemma 4 (identities)**.** *With F_t the isolated map,*

$$u' = F_t(u) + \frac{w_2}{D_2}\Big(v + \frac{w_1 u}{D_1}\Big), \qquad v' = F_t(v) + \frac{w_1}{D_2}\Big(u + \frac{w_2 v}{D_3}\Big), \tag{13}$$

$$h' = h - w_1\Big[u\Big(\frac{1}{2D_2} + \frac{1}{D_1}\Big) + \frac{w_2 v}{2D_2 D_3}\Big], \quad \tilde{h}' = \tilde{h} - w_2\Big[v\Big(\frac{1}{2D_2} + \frac{1}{D_3}\Big) + \frac{w_1 u}{2D_1 D_2}\Big], \tag{14}$$

$$D' = D - \frac{w_1 u}{D_1} - \frac{w_2 v}{D_3}. \tag{15}$$

Consequently $h, \tilde{h}, D$ are nonincreasing, $u \leq h$, $v \leq \tilde{h}$, $u, v \leq D \leq D_0$, and

$$(1 - 2w_1)u \leq u' \leq (1 - w_1)u + w_2(u + v), \qquad (1 - 2w_2)v \leq v' \leq (1 - w_2)v + w_1(u + v), \tag{16}$$

$$0 \leq h - h' \leq \tfrac{3}{2}\, w_1(u + v). \tag{17}$$

Proof. Put $x_1 = 0$, $x_2 = u$, $x_3 = u + v$. Then $x_1' = w_1 u/D_1$ and $x_2' = (u(1 + w_2) + w_2 v)/D_2$, so $u' = u\frac{1+w_2-w_1^2}{D_1 D_2} + \frac{w_2 v}{D_2}$. Since $\frac{1+w_2-w_1^2}{D_1 D_2} - \frac{1-w_1}{1+w_1} = \frac{w_1 w_2}{D_1 D_2}$, this is the first identity in (13); the second follows by the mirror symmetry $x \mapsto -x$, $1 \leftrightarrow 3$. With $\Delta x_1 = w_1 u/D_1$, $\Delta x_2 = (w_2 v - w_1 u)/D_2$ and $\Delta x_3 = -w_2 v/D_3$ we get $h' - h = \frac{\Delta x_2 + \Delta x_3}{2} - \Delta x_1$, and $\frac{1}{D_2} - \frac{1}{D_3} = -\frac{w_1}{D_2 D_3}$ gives (14); the mirror gives the $\tilde{h}$ identity. (15) is $\Delta x_3 - \Delta x_1$. The bounds use $1 - 2a \leq \frac{1-a}{1+a} \leq 1 - a$, $D_i \geq 1$, $w_i \leq 1$, and, for (17), $\frac{1}{2D_2} \leq \frac{1}{2}$, $\frac{1}{D_1} \leq 1$ and $\frac{w_2}{D_2 D_3} \leq \frac{1}{4}$. □

(The identities were checked against the implementation on $2 \cdot 10^5$ random points; the maximal discrepancy is $2.7 \cdot 10^{-15}$.)

Proof of Theorem 3(a). If $h(t_1) < 1$ for some t_1, then $u_\infty \leq h_\infty \leq h(t_1) < 1$, so $u_\infty = 0$ by the dichotomy; at $t_1 = 0$ this is the first claim. Let $u_0 + v_0 < 2$. If gap 1 persists, then $v_\infty = 0$: otherwise $v_\infty \geq 1$ and $u_\infty \geq 1$ give $D_\infty \geq 2 > D_0$, contradicting (15). Hence $u_\infty = h_\infty$ and $h(t) \geq h_\infty \geq 1$ for all t. Conversely, if $h(t) \geq 1$ for all t but $u_\infty = 0$, then $h_\infty = v_\infty/2 \leq D_0/2 < 1$, a contradiction. On the symmetric line the dynamics commute with the mirror, so $u(t) = v(t)$; if gap 1 persisted, the previous argument would force $v_\infty = 0 \neq u_\infty$. □

Proof of Theorem 3(b). Let $L = \log t_0$, $w_* = t_0^{-v_0^2}$, $p_0 = \frac{w_* v_0}{1 + w_*}$, $d_- = 1 - p_0$, and

$$p := \frac{w_* v_0}{1 + t_0^{-d_-^2} + w_*}, \qquad m := \max\{d_-^2, \tfrac{3}{2L}\}, \qquad M := 1 + (4mL - 2)_+\, t_0^{-m}, \qquad R := p/M.$$

First, $p_0 \leq \sup_v v e^{-v^2 L} = (2eL)^{-1/2} < 1$ for $L \geq \log 2$, so $d_- > 0$, and $R \leq p \leq p_0 < \kappa(t_0) - d_-$. For $u_0 \geq \kappa(t_0)$ use Theorem 2(a). For $u_0 \in [\kappa - R, \kappa)$ we have $u_0 > d_-$, and by (13) at $t = 0$, $u(1) \geq F_0(u_0) + \frac{w_* v_0}{1 + t_0^{-u_0^2} + w_*} \geq F_0(u_0) + p$. By the mean value theorem, $F_0(\kappa) - F_0(u_0) \leq (\kappa - u_0) \sup_{d \geq d_-} F_0'(d)$. Moreover $F_0'(d) - 1 = \frac{a(4d^2 L - 2 - 2a)}{(1+a)^2} \leq G(d^2)$ with $G(m') := (4m'L - 2)_+ e^{-m'L}$. G vanishes on $[0, \frac{1}{2L}]$, increases on $[\frac{1}{2L}, \frac{3}{2L}]$ and decreases afterwards, so $\sup_{m' \geq d_-^2} G(m') = G(m)$. Hence $F_0(\kappa) - F_0(u_0) \leq RM = p$ and $u(1) \geq F_0(\kappa(t_0)) = \kappa(t_0 + 1)$ by Theorem 1(iv). Persistence follows from Theorem 2(a). □

Remark 4 (size of the one-step rescue). As $t_0 \to \infty$ with $v_0 = x/\sqrt{L}$: $\sup_{v_0} R = (c_1 + o(1))/\sqrt{\log t_0}$ with $c_1 = \max_x xe^{-x^2}/(1+e^{-x^2}) = 0.2779$. At $t_0 = 5, 11, 100, 10^3, 10^6$ the largest one-step rescue $\max_{v_0} R/\kappa$ is $5.5\%, 5.5\%, 6.8\%, 7.5\%, 6.2\%$. The macroscopic rescue (25–33%) needs $\gtrsim t_0^{v_0^2}$ steps, which is the content of part (c).

Scaling limit. The three window lemmas below hold for $t_0 \geq 2$ and an integer horizon $T \leq t_0$, so that $t_0 \leq s_t \leq 2t_0$ for $t \leq T$.

Lemma 5 (persistence window). *Let $0 < b < a$, $\epsilon := Tt_0^{-a^2}$ and $E := \exp(-T(2t_0)^{-b^2})$. If $u_0 \geq a + 2D_0\epsilon$ and $v_0 + D_0\epsilon \leq b$, then $u(T) \geq u_0 + \frac{v_0}{2} - 2D_0\epsilon - \frac{v_0}{2}E$. Consequently gap 1 persists if $u_0 + \frac{v_0}{2} - 2D_0\epsilon - \frac{v_0}{2}E \geq \kappa(t_0)$.*

Proof. By induction on $t \leq T$: $u(t) \geq u_0 - 2D_0 t\, t_0^{-a^2} \geq a$ and $v(t) \leq v_0 + D_0 t\, t_0^{-a^2} \leq b$. Indeed, if $u(t) \geq a$ then $w_1(t) \leq t_0^{-a^2}$, and (16) with $u, v \leq D_0$ gives $u(t+1) \geq u(t) - 2D_0 t_0^{-a^2}$ and $v(t+1) \leq v(t) + D_0 t_0^{-a^2}$. Then $w_2(t) \geq (2t_0)^{-b^2}$, so $v(t+1) \leq (1-(2t_0)^{-b^2})v(t) + D_0 t_0^{-a^2}$ and $v(T) \leq v_0 E + D_0\epsilon$. By (17), $h(T) \geq h_0 - \frac{3}{2}D_0\epsilon$. Hence $u(T) = h(T) - v(T)/2$ satisfies the bound. Finally $\kappa(t_0+T) \leq \kappa(t_0)$ and Theorem 2(a) applies. □

Lemma 6 (race window). *Let $0 < \bar{u} < v_{\mathrm{lo}}$, $\epsilon' := Tt_0^{-v_{\mathrm{lo}}^2}$ and $E' := \exp(-T(2t_0)^{-\bar{u}^2})$. If $u_0 + D_0\epsilon' \leq \bar{u}$ and $v_0(1-2\epsilon') \geq v_{\mathrm{lo}}$, then $u(T) \leq u_0 E' + D_0\epsilon'$. Hence gap 1 merges if $u_0 E' + D_0\epsilon' + \frac{v_0}{2} + \frac{u_0}{4} < 1$.*

Proof. By induction on $t \leq T$: $v(t) \geq v_0(1-2t_0^{-v_{\mathrm{lo}}^2})^t \geq v_0(1-2\epsilon') \geq v_{\mathrm{lo}}$, by the lower bound in (16) with $w_2 \leq t_0^{-v_{\mathrm{lo}}^2}$. Also $u(t) \leq u_0 + D_0 t\, t_0^{-v_{\mathrm{lo}}^2} \leq \bar{u}$, by the upper bound in (16). Then $w_1(t) \geq (2t_0)^{-\bar{u}^2}$, so $u(t+1) \leq (1-(2t_0)^{-\bar{u}^2})u(t) + D_0 t_0^{-v_{\mathrm{lo}}^2}$, which gives the bound on $u(T)$. Moreover $v(T) \leq \tilde{h}(T) \leq \tilde{h}_0 = v_0 + u_0/2$. Hence $h(T) < 1$, and part (a) applies. □

Lemma 7 (persistent neighbour). *Let $v_{\mathrm{lo}} > 1$ and $P_\infty := (t_0-1)^{1-v_{\mathrm{lo}}^2}/(v_{\mathrm{lo}}^2 - 1)$. If $v_0(1-2P_\infty) \geq v_{\mathrm{lo}}$ and $u_0 + (v_{\mathrm{lo}}+1)P_\infty < 1$, then gap 1 merges and gap 2 persists.*

Proof. By induction, $v(t) \geq v_0 \prod_{u'<t}(1 - 2s_{u'}^{-v_{\mathrm{lo}}^2}) \geq v_0(1-2P_\infty) \geq v_{\mathrm{lo}}$. The push in (13) is at most $w_2 v + w_2 u \leq (v_{\mathrm{lo}}+1)s_t^{-v_{\mathrm{lo}}^2}$, because $x \mapsto xs^{-x^2}$ decreases for $x \geq 1/\sqrt{2\log s}$ and $u < 1$, and $F_t(u) \leq u$. Hence $u_\infty \leq u_0 + (v_{\mathrm{lo}}+1)P_\infty < 1$, so $u_\infty = 0$. □

Theorem 7 (explicit form of Theorem 3(c)). *Let $x \in (0,1)\cup(1,\infty)$, $y > 0$, $y \neq \varphi(x)$. There is an explicit $T^*(x,y)$ such that for all $t_0 \geq T^*$ the chain started from $(u_0, v_0) = (y\kappa(t_0), x\kappa(t_0))$ has gap 1 persisting if $y > \varphi(x)$ and merging if $y < \varphi(x)$. With $\bar{\kappa}$ any certified upper bound for $\kappa(T^*)$ (for instance from C1) and $T = \lceil t_0^\beta \rceil$, the following conditions at $t_0 = T^*$ suffice; all of them get weaker as t_0 grows.*

- *(i) ($x < 1$, $\varphi(x) < y < 1$; persistence) $x\bar{\kappa} < b < \sqrt{\beta} < a < y$, $\beta < 1$, $t_0^\beta + 1 \leq t_0$, and with $e = 2t_0^{\beta-a^2}$: $y - 2(x+1)\bar{\kappa}e \geq a$, $x\bar{\kappa} + (x+1)\bar{\kappa}e \leq b$, and $y + \frac{x}{2} - 1 \geq 2(x+1)e + \frac{x}{2}\exp(-2^{-b^2}t_0^{\beta-b^2})$.*
- *(ii) ($y < 1 - x/2$, $x < 2$; merge) $\bar{\kappa}\,(y + x/2) < 1$.*
- *(iii) ($2/3 < x < 1$, $1 - x/2 \leq y < x$; merge) $y\bar{\kappa} < \bar{u} < \sqrt{\beta} < v_{\mathrm{lo}} < x$, $t_0^\beta + 1 \leq t_0$, $e' = 2t_0^{\beta - v_{\mathrm{lo}}^2}$, $y\bar{\kappa} + (x+y)\bar{\kappa}e' \leq \bar{u}$, $x(1-2e') \geq v_{\mathrm{lo}}$, and $y\bar{\kappa}e^{-2^{-\bar{u}^2}t_0^{\beta-\bar{u}^2}} + (x+y)\bar{\kappa}e' + \bar{\kappa}(\frac{x}{2} + \frac{y}{4}) < 1$.*
- *(iv) ($x > 1$, $y < 1$; merge) $1 < v_{\mathrm{lo}} < x$ with $x(1-2P_\infty) \geq v_{\mathrm{lo}}$ and $y\bar{\kappa} + (v_{\mathrm{lo}}+1)P_\infty < 1$.*
- *(v) ($y \geq 1$) persistence for every t_0, by Theorem 2(a).*

Each condition holds for all large t_0, because $\bar{\kappa}$ can be taken $\to 1$ by Theorem 1(v), $t_0^{\beta-a^2}, t_0^{\beta-v_{\mathrm{lo}}^2}, P_\infty \to 0$ and $t_0^{\beta-b^2}, t_0^{\beta-\bar{u}^2} \to \infty$. The regions (i)–(v) cover all (x,y) with $x \neq 1$ and $y \neq \varphi(x)$.

Proof. (i) Apply Lemma 5 with these a, b, T, using $\kappa \geq 1$ (so $u_0 \geq y$), $\kappa \leq \bar{\kappa}$, $D_0 = (x+y)\kappa \leq (x+1)\bar{\kappa}$ and $t_0^\beta \leq T \leq 2t_0^\beta$. The final inequality of Lemma 5, divided by κ, reads $y + x/2 - 1 \geq 2(x+y)\epsilon + \frac{x}{2}E$. Such a, b exist because $y > x$, and the left side is positive because $y > 1 - x/2$. (ii) $u_0 + v_0/2 = \kappa(y + x/2) < 1$; use part (a). (iii) Lemma 6 with $D_0 \leq (x+y)\bar{\kappa}$, $v_0 \geq x$, $u_0 \leq y\bar{\kappa}$ and $v_0/2 + u_0/4 \leq \bar{\kappa}(x/2 + y/4)$. As $t_0 \to \infty$ the left side of the last condition tends to $x/2 + y/4 < 3/4$. (iv) Lemma 7 with $v_0 \geq x$ and $u_0 \leq y\bar{\kappa}$. Coverage: for $x \leq 2/3$, $\varphi = 1 - x/2$ and $y < \varphi$ is (ii); for $2/3 < x < 1$, $\varphi = x$ and $y < \varphi$ is (ii)$\cup$(iii); for $x > 1$, $y < 1$ is (iv); and $y > \varphi$ is (i) or (v). □

Proof of Theorem 3(c). Pointwise: within each region the conditions of Theorem 7 are monotone in y (larger y only helps (i), smaller y only helps (ii)–(iv); in (i) use $x + 1 \geq x + y'$). So for each $\epsilon > 0$, eventually gap 1 persists for all $y' \geq \varphi(x) + \epsilon$ and merges for all $y' \leq \varphi(x) - \epsilon$. For the infimum, $\limsup \inf_{v_0} U_+/\kappa \leq \lim U_+(\frac{2}{3}\kappa)/\kappa = \frac{2}{3}$. For the lower bound, fix $\epsilon \in (0, \frac{1}{6})$ and $y \leq \frac{2}{3} - \epsilon$; we show merging uniformly in $x > 0$ for large t_0. If $x \leq \frac{2}{3} + \epsilon$, then $y + x/2 \leq 1 - \epsilon/2$ and (ii) holds once $\bar{\kappa} < 1/(1 - \epsilon/2)$. If $\frac{2}{3} + \epsilon \leq x \leq \frac{4}{3}$, apply Lemma 6 with $\bar{u} = \frac{2}{3} - \frac{\epsilon}{2}$, $v_{\mathrm{lo}} = \frac{2}{3} + \frac{\epsilon}{2}$, $\beta = (\bar{u}^2 + v_{\mathrm{lo}}^2)/2$ and $D_0 \leq 2\kappa$; the last condition becomes $\bar{\kappa}(\frac{x}{2} + \frac{y}{4}) \leq \frac{5}{6}\bar{\kappa} < 1$ up to vanishing terms, uniformly in x. If $x \geq \frac{4}{3}$, apply Lemma 7 with $v_{\mathrm{lo}} = \frac{7}{6}$, uniformly in x. Hence $U_- \geq (\frac{2}{3} - \epsilon)\kappa$ for all v_0 once t_0 is large. □

Remark 5 (finite t_0 and the size of T^*). A sharper finite-t_0 check (merge iff $h < 1$; persist iff u exceeds the critical trajectory from the upper end of the κ enclosure) at $t_0 = 10^5$ gives $u^*/\kappa = 0.937, 0.837, 0.737, 0.692, 0.684, 0.703, 0.800, 0.900$ for $x = 0.1, 0.3, 0.5, 0.6, \frac{2}{3}, 0.7, 0.8, 0.9$, against $\varphi = 0.950, 0.850, 0.750, 0.700, 0.667, 0.700, 0.800, 0.900$. On the branch $\varphi = x$ the agreement is exact already at finite t_0, as the symmetric line predicts. On the branch $1 - x/2$ the finite-t_0 boundary lies slightly below φ. The constants in Theorem 7 are explicit but crude: uniform persistence for $x \in [0.3, 0.5]$, $y \geq 0.9$ holds from $t_0 \geq 2.0 \cdot 10^8$, but near the corner $x = y = 2/3$ the bound is $T^* \approx 5 \cdot 10^{110}$, because the proof separates the time scales $t_0^{v_0^2} \ll t_0^{u_0^2}$. For moderate t_0 the macroscopic rescue is established by Theorem 4.

C Proofs for general graphs

Lemma 8 (32, Thm. 1, deterministic case)**.** *Let $A(t)$ be row-stochastic $n \times n$ matrices, and suppose there are $\alpha, \gamma > 0$ such that for all t: (i) $A_{ii}(t) \geq \gamma$ for all i; (ii) $\sum_{i \in S, j \notin S} A_{ij}(t) \geq \alpha \sum_{i \notin S, j \in S} A_{ij}(t)$ for every nonempty proper $S \subset [n]$. Then $x(t) = A(t-1) \cdots A(0)x(0)$ converges for every $x(0)$, and $\lim_t x_i(t) = \lim_t x_j(t)$ whenever i, j lie in the same connected component of the infinite flow graph $\{\{i, j\} : \sum_t (A_{ij}(t) + A_{ji}(t)) = \infty\}$.*

(Touri and Nedić state the theorem for independent random chains; a deterministic chain is a special case. Their weak aperiodicity follows from (i), since $\sum_l A_{li}A_{lj} \geq \gamma(A_{ij} + A_{ji})$. The discrete-time cut-balanced theorem of Hendrickx and Tsitsiklis [16] additionally needs a lower bound on the positive entries, which fails here because attention weights decay to 0.)

Proof of Proposition 3. For a fixed initial condition the matrices $A_{ij}(t) = \alpha_{ij}(t)$ form a fixed deterministic sequence. By (2), $A_{ii} = 1/\pi_i \geq 1/N_{\max}$. Since w is symmetric, $\sum_{i \in S, j \notin S} \frac{w_{ij}}{\pi_i} \geq \frac{1}{N_{\max}} \sum_{i \in S, j \notin S} w_{ij} = \frac{1}{N_{\max}} \sum_{i \notin S, j \in S} w_{ij} \geq \frac{1}{N_{\max}} \sum_{i \notin S, j \in S} \frac{w_{ij}}{\pi_i}$, so (ii) of Lemma 8 holds with $\alpha = 1/N_{\max}$. Apply Lemma 8 to each coordinate; $A_{ij} + A_{ji} \geq 2w_{ij}/N_{\max}$ gives (i) and (ii) of the proposition. For (iii), if $\ell_{ij} \in (0, 1)$, pick $\ell' \in (\ell_{ij}, 1)$; eventually $\|z_i - z_j\| < \ell'$, so $w_{ij}(t) \geq s_t^{-\ell'^2}$ and $\sum_t w_{ij} = \infty$, which contradicts (ii). □

Proof of Theorem 5. Let $\varepsilon_t := e_{\max}\delta s_t^{-\delta^2}$ and $\eta_t := \sum_{u \leq t} \varepsilon_u \leq \eta$, by the integral bound and (A3). We prove by induction $H(t)$: $z_i(t) \in C_k + \bar{B}(\eta_t)$ for all k and $i \in B_k$. $H(0)$ holds. Assume $H(t)$. If $i \in B_k$, $j \in B_l$, $k \neq l$ and $\{i, j\} \in E$, then $k \sim l$ and $\|z_i - z_j\| \geq$

$\operatorname{dist}(C_k, C_l) - 2\eta_t \geq \delta$ by (A2). For $i \in B_k$ write $z_i(t+1) = y_i + \xi_i$ with

$$y_i := z_i + \sum_{j \in N(i) \cap B_k} \alpha_{ij}(z_j - z_i), \qquad \xi_i := \sum_{j \in N(i) \setminus B_k} \alpha_{ij}(z_j - z_i).$$

The point y_i is a convex combination of $\{z_j(t) : j \in B_k\}$: the coefficient of z_i is $\alpha_{ii} + \sum_{j \notin B_k} \alpha_{ij} \geq 0$, and the coefficients sum to 1. Hence $y_i \in C_k + \bar{B}(\eta_t)$. Since $\pi_i \geq 1$, $d_{ij} \geq \delta$ and $x \mapsto x e^{-x^2 L}$ decreases on $[1/\sqrt{2L}, \infty)$ with $\delta > 1 \geq 1/\sqrt{2 \log 2}$, we get $\|\xi_i\| \leq \sum_{j \in N(i) \setminus B_k} d_{ij} e^{-d_{ij}^2 L_t} \leq \varepsilon_t$. This is $H(t+1)$. For a within-block edge, $\|z_i - z_j\| \leq D_k + 2\eta \leq 1$ by (A1), so $w_{ij} \geq s_t^{-1}$ and $\sum_t w_{ij} = \infty$; by Proposition 3(ii) and connectivity of $G[B_k]$, all of B_k shares one limit in the closed set $C_k + \bar{B}(\eta)$. Adjacent block limits are at distance $\geq \operatorname{dist}(C_k, C_l) - 2\eta \geq \delta$. □

Proof of Corollary 2. (a) Take $C_k = \bar{B}(\mu_k, r_k)$, so $D_k = 2r_k$ and $\operatorname{dist}(C_k, C_l) \geq \|\mu_k - \mu_l\| - r_k - r_l$. (b) Let $\ell := \log(t_0 - 1) > e$, $\gamma := 2 \log \ell / \ell \in (0, 1)$, $\delta := \sqrt{1+\gamma}$, $\eta := e_{\max}\delta/(2\ell \log \ell)$, and $\Gamma(t_0) := \delta + 2\eta = \sqrt{1+\gamma}\,(1 + \frac{e_{\max}}{\ell \log \ell})$. Then $(t_0 - 1)^{1-\delta^2} = e^{-\gamma \ell} = \ell^{-2}$, so (A3) holds with equality. (A1) holds once $2\eta \leq \theta$, and (A2) is the separation assumption. Finally $\sqrt{1+\gamma} \leq 1 + \log \ell / \ell$ and $\ell \sim \log t_0$. Necessity: for a block-regular graph with block-constant features, separation below $\kappa(t_0; \rho_A, \rho_B)$ merges the blocks by Theorem 1(ii). □

Proof of Proposition 4. Take two k-cliques Q_1, Q_2 joined by a perfect matching, with $z \equiv 0$ on Q_1 and $z \equiv D$ on Q_2. This is block-regular with $\rho_A = \rho_B = 1/k$, and the gap obeys $d' = d\,\frac{k-a}{k+a} \geq d(1 - 2a/k)$ with $a = s_t^{-d^2}$. Let $\delta := (1 + D)/2$. If $d_u \geq \delta$ for all $u < t$, then $d_t \geq D\big(1 - \frac{2}{k}\frac{(t_0-1)^{1-\delta^2}}{\delta^2 - 1}\big) \geq \delta$, provided $k \geq \frac{2D(t_0-1)^{1-\delta^2}}{(\delta^2-1)(D-\delta)}$. By induction the gap stays $\geq \delta > 1$. (At $t_0 = 602$, $k = 100$ and $D = 1.1$, the gap ends at 1.071.) □

D Proofs for learned attention

Throughout, $c = 1$, f is a score, $m_{ij} = f(z_i, z_i) - f(z_i, z_j)$ and $w_{ij} = s_t^{-m_{ij}}$.

Proof of Proposition 5. Write $x = z_i$, $y = z_j$ and $J = J_s + J_a$. The terms of $f(x, \cdot)$ that depend only on x cancel, and $x^\top J_a x = 0$, so $m_{ij} = x^\top J_s x - x^\top J_s y - x^\top J_a y - \frac{1}{2} x^\top \Omega x + \frac{1}{2} y^\top \Omega y + b^\top (x - y)$. Expanding $\frac{1}{2}(x-y)^\top J_s (x-y) + \mathcal{V}(x) - \mathcal{V}(y)$ gives the same expression without the term $-x^\top J_a y$. Adding the formula for m_{ji}, the potential terms cancel and $x^\top J_a y + y^\top J_a x = 0$. □

Lemma 9 (two-block reduction, any score)**.** *On a block-regular graph with block-constant features, $z'_A = z_A + p_A(z_B - z_A)$ and $z'_B = z_B - p_B(z_B - z_A)$ with $p_X = \rho_X a_X/(1 + \rho_X a_X)$, $a_A = s_t^{-m_{AB}}$ and $a_B = s_t^{-m_{BA}}$. Hence $z'_B - z'_A = k(z_B - z_A)$ with $k = 1 - p_A - p_B \in (-1, 1)$, and both new values lie on the segment between the old ones.*

Proof. A node of A sees k_A nodes with margin 0 and e_A nodes with margin m_{AB}, so $z'_A = (k_A z_A + e_A a_A z_B)/(k_A + e_A a_A)$. The same holds for every node of A, and symmetrically for B. Since $p_X \in [0, 1)$, $|k| < 1$, and expanding $1 - p_A - p_B$ over the common denominator gives $k = (1 - P a_A a_B)/((1 + \rho_A a_A)(1 + \rho_B a_B))$. □

The margin map (7). By Lemma 9 the block values stay on the line $\{p + \xi e\}$ through the initial values. For $x = p + \xi e$ and $y = p + \eta e$ we have $x^\top J y = \sigma \xi \eta + \xi\, e^\top J p + \eta\, p^\top J e + p^\top J p$, because $e^\top J_a e = 0$, and $-\frac{1}{2} y^\top \Omega y + b^\top y = -\frac{1}{2}\sigma\omega\eta^2 + \eta(b - \Omega p)^\top e + \text{const}$. With $\tau = p^\top J e + (b - \Omega p)^\top e$, the terms in ξ alone cancel and

$$f(x, x) - f(x, y) = \tfrac{1}{2}\sigma\omega\Delta^2 - \Delta\, g(\xi), \qquad \Delta = \eta - \xi, \qquad g(\xi) = \sigma(1 - \omega)\xi + \tau.$$

Let $\xi_A < \xi_B$ be the block positions and $d = \xi_B - \xi_A$ (signed, so that crossings are allowed). Then $M_A = \frac{1}{2}\sigma\omega d^2 - d\, g(\xi_A)$, $M_B = \frac{1}{2}\sigma\omega d^2 + d\, g(\xi_B)$ and, since $g(\xi_B) - g(\xi_A) = \sigma(1 - \omega)d$,

$M_A + M_B = \sigma d^2$. By Lemma 9, $\xi'_A = \xi_A + p_A d$, $d' = kd$ and $g(\xi'_A) = g(\xi_A) + \sigma(1-\omega)p_A d$. Substituting $d\,g(\xi_A) = \frac{1}{2}\sigma\omega d^2 - M_A$,

$$M'_A = \tfrac{1}{2}\sigma\omega k^2 d^2 - kd\,g(\xi'_A) = k\big[M_A - \sigma d^2\big(\tfrac{\omega}{2}(1-k) + (1-\omega)p_A\big)\big],$$

which is (7) because $1-k = p_A + p_B$ and $\sigma d^2 = M_A + M_B$. The case of B is symmetric. We checked the map against the feature-space dynamics for random J, Ω, b in $\mathbb{R}^3$ (relative error below 10^{-13}).

Lemma 10 (two properties of the map)**.** *Let $\sigma > 0$, $0 \le \omega \le 1$ and $P \le 1$. Then $0 < k < 1$ at every layer, and:*

(a) if $M_X > 0$, then $M'_X \le M_X$;

(b) if $M_X \le 0$, then $M'_X \le 0$ and $k \le 1/(1+\rho_X)$.

Proof. $a_A a_B = s_t^{-(M_A+M_B)} = s_t^{-\sigma d^2} < 1 \le 1/P$, so $k > 0$ by Lemma 9, and $k < 1$ because $p_A, p_B > 0$. The bracket in (7) is at most M_X, because $\frac{\omega}{2}(p_A+p_B) + (1-\omega)p_X \ge 0$. (a) If the bracket is nonnegative, $M'_X \le$ bracket $\le M_X$; otherwise $M'_X < 0 < M_X$. (b) The bracket is at most $M_X \le 0$, so $M'_X \le 0$. Moreover $a_X \ge 1$ gives $p_X \ge \rho_X/(1+\rho_X)$ and $k \le 1 - p_X \le 1/(1+\rho_X)$. □

Proof of Theorem 6. (i) Since $0 < k < 1$, the blocks never cross and each block value moves monotonically towards the other, so both converge and d_t decreases. Suppose first that $\min(M_A, M_B) > 1$ at every layer. By Lemma 10(a) the margins decrease to limits ≥ 1, and the limiting gap satisfies $\sigma d_\infty^2 = \lim(M_A + M_B) \ge 2$.

Suppose now that $\min(M_A, M_B) \le 1$ at layer T, say $M_X(T) \le 1$. By Lemma 10, $M_X(t) \le 1$ for all $t \ge T$: a positive margin does not increase, and a nonpositive one stays nonpositive. Hence $a_X \ge s_t^{-1}$, $p_X \ge \frac{\rho_X}{1+\rho_X} s_t^{-1}$ and $d_{t+1} = k\,d_t \le (1-p_X)d_t \le \big(1 - \frac{\rho_X}{1+\rho_X} s_t^{-1}\big)d_t$ for $t \ge T$. Since $\sum_t s_t^{-1} = \infty$, $d_t \to 0$, and the two block values converge to one point.

(ii) If $M_A = M_B$ and $\rho_A = \rho_B = \rho$, then $p_A = p_B =: p$, the two brackets in (7) agree, and $M'_X = k(M_X - 2M_X p) = k^2 M_X$. Here $k = 1 - 2p = (1-\rho a)/(1+\rho a) = \psi(a)$ with $a = s_t^{-M_X} = s_t^{-(\sqrt{M_X})^2}$. So $\sqrt{M'_X} = F_t(\sqrt{M_X})$ with F_t from (3) for $\rho_A = \rho_B = \rho$.

(iii), necessity. Suppose the pair persists. By (i) both margins exceed 1 at all layers, so (a) of Lemma 10 applies throughout. Let X be a block with $M_X(0) = \min(M_A, M_B) = 1 + \varepsilon$. Then $1 < M_X(t) \le 1 + \varepsilon$, hence $s_t^{-(1+\varepsilon)} \le a_X \le s_t^{-1} \le t_0^{-1}$ and $p_X \ge \rho_X(1+\rho_X/t_0)^{-1} s_t^{-(1+\varepsilon)}$. Dropping $\frac{\omega}{2} p_Y \ge 0$ in the bracket and using $k \le 1$ and $M_A + M_B \ge M_X$, $M'_X \le M_X\big(1 - (1-\frac{\omega}{2})p_X\big)$. Therefore

$$\varepsilon \;\ge\; \log\frac{M_X(0)}{M_X(\infty)} \;\ge\; \Big(1 - \frac{\omega}{2}\Big)\sum_{t\ge 0} p_X(t) \;\ge\; \Big(1 - \frac{\omega}{2}\Big)\rho_-\,\frac{t_0^{-\varepsilon}}{\varepsilon},$$

using $\sum_{t\ge 0}(t_0+t)^{-1-\varepsilon} \ge t_0^{-\varepsilon}/\varepsilon$. So $\varepsilon^2 \ge (1-\frac{\omega}{2})\rho_- t_0^{-\varepsilon}$, and $\varepsilon \ge \varepsilon_\omega$ because the left side increases and the right side decreases in ε.

(iii), certificate C1′. Suppose both margins are at least δ^2 at all layers $u < t$. They do not increase (Lemma 10(a)), so $M_A + M_B \le R\delta^2$ and $(M_A + M_B)/M_X \le R$. Moreover $p_X \le \rho_X s_u^{-\delta^2} =: \rho_X q_u$, so $p_A + p_B \le \lambda q_u$ and $\frac{\omega}{2}(p_A+p_B) + (1-\omega)p_X \le \lambda q_u$. From (7),

$$\log M'_X \ge \log M_X + \log(1 - \lambda q_u) + \log(1 - R\lambda q_u) \ge \log M_X - \frac{(1+R)\lambda q_u}{1 - R\lambda q},$$

using $R \ge 2$ and $q_u \le q$. Summing with $\sum_{u\ge 0} q_u \le (t_0 - 1)^{1-\delta^2}/(\delta^2 - 1)$ shows $\log M_X(t) \ge \log m - \log(m/\delta^2) = \log \delta^2$, and induction on t closes the argument.

(iv) Take $\gamma_- = \varepsilon_\omega$. As $L = \log t_0 \to \infty$, $\varepsilon_\omega L = \log((1-\frac{\omega}{2})\rho_-) - 2\log\varepsilon_\omega$, so $\varepsilon_\omega = (2 + o(1))\log L/L$. For γ_+, let $\ell = \log(t_0 - 1)$ and $\delta^2 = 1 + 2\log\ell/\ell$. Then $(t_0-1)^{1-\delta^2}/(\delta^2 - 1) =$

$1/(2\ell\log\ell)$, $R \le \bar{M}$ and $q \to 0$. So C1′ holds as soon as $\log(m/\delta^2) \ge (1+o(1))(1+\bar{M})\lambda/(2\ell\log\ell)$, which is implied by $m \ge 1+\gamma_+$ with $\gamma_+ = 2\log\ell/\ell + O(1/(\ell\log\ell))$. By Theorem 1(v), $\kappa(t_0)^2 = 1+(2+o(1))\log L/L$ as well. □

Proof of Proposition 6. Let $z_A = -v$ and $z_B = v$. Then $m_{AB} = (-v)^\top J(-2v) = 2v^\top J_s v = m_{BA}$, so $p_A = p_B = p$, and by Lemma 9 the midpoint stays at 0. Hence $z_A = -z_B$ for all t. With $d = \|z_B - z_A\| = 2\|v\|$ and $D^2 = |\sigma|d^2/2$, both margins equal $-D^2$. So $a = s_t^{D^2} \ge 1$, and $\rho a \ge 1$ gives $k = 1-2p = -(\rho a-1)/(\rho a+1) \le 0$, with $k<0$ when $D>0$. Writing $\tilde{a} = 1/a = s_t^{-D^2}$,

$$|k| = \frac{1-\tilde{a}/\rho}{1+\tilde{a}/\rho} = \frac{1-\tilde{a}^2/\rho^2}{(1+\tilde{a}/\rho)^2},$$

which is $\psi(\tilde{a})$ of Lemma 1 for cut ratios $(1/\rho, 1/\rho)$, with $P = 1/\rho^2 \le 1$. So $D_{t+1} = F_t(D_t)$ for these cut ratios, and Theorem 1(i)–(ii) gives the dichotomy. In the persisting case $k<0$ at every layer, so $z_B - z_A$ changes sign at every layer, while $\sum_t \alpha_{AB}(t) = \infty$ because $p \ge \frac{1}{2}$. □

Remark 6 (GAT). Let $f(x,y) = \phi(u^\top x + v^\top y)$ with $\phi(r) = \max(r, \beta r)$, $0 \le \beta < 1$, and write $U_i = u^\top z_i$ and $V_i = v^\top z_i$. Since $\phi(r) = \beta r + (1-\beta)r_+$ and the linear parts cancel,

$$m_{ij} + m_{ji} = (1-\beta)\big[(U_i+V_i)_+ - (U_i+V_j)_+ - (U_j+V_i)_+ + (U_j+V_j)_+\big].$$

Grouping the four terms in pairs by U, or by V, and using that $r \mapsto r_+$ is nondecreasing and 1-Lipschitz, gives $|m_{ij}+m_{ji}| \le (1-\beta)\min(|U_i-U_j|, |V_i-V_j|)$. If the four arguments U_i+V_i, U_i+V_j, U_j+V_i and U_j+V_j share a sign, ϕ is linear on them and $m_{ij} = -m_{ji}$. On a two-block pair one block X then has a nonpositive margin, $a_A a_B = 1$, and by Lemma 9 $0 \le k \le 1-p_X \le 1/(1+\rho_X)$.

Proposition 7 (separation for any score)**.** *Let $V = B_1 \sqcup \cdots \sqcup B_K$, let $C_k \supseteq \{z_i(0) : i \in B_k\}$ be convex, let $\eta \ge 0$ and $\delta > 1$, and let $C_k^\eta = C_k + \bar{B}(\eta)$. Suppose that $f(x,x) - f(x,y) \ge \delta^2$ for all $x \in C_k^\eta$, $y \in C_l^\eta$ and $k \sim l$, and that $e_{\max} R\,(t_0-1)^{1-\delta^2}/(\delta^2-1) \le \eta$, where $R = \max_{k\sim l} \operatorname{diam}(C_k^\eta \cup C_l^\eta)$. Then $z_i(t) \in C_k^\eta$ for all t and $i \in B_k$. In particular no node of B_k comes closer than $\operatorname{dist}(C_k, C_l) - 2\eta$ to a node of an adjacent block B_l.*

Proof. As in the proof of Theorem 5, with $\eta_t = \sum_{u<t} e_{\max} R s_u^{-\delta^2} \le \eta$, assume $z_i(t) \in C_k + \bar{B}(\eta_t)$ for all k and $i \in B_k$, and split $z_i(t+1) = y_i + \xi_i$ into the in-block part y_i, a convex combination of in-block values, and the cross-block part $\xi_i = \sum_{j \in N(i)\setminus B_k} \alpha_{ij}(z_j - z_i)$. Since $\pi_i \ge w_{ii} = 1$, each cross weight is $\alpha_{ij} \le w_{ij} \le s_t^{-\delta^2}$, and $\|z_j - z_i\| \le R$. Hence $\|\xi_i\| \le e_{\max} R s_t^{-\delta^2}$, which is the induction step. The integral bound gives $\eta_t \le \eta$. □

E Validated numerics

Interval evaluation. For the two-block map we propagate an interval $[d^-, d^+]$ using monotonicity: F_t is increasing in d, so $[F_t(d^-), F_t(d^+)]$ encloses the image, evaluated with outward rounding. For chains we evaluate (4) with interval weights $w_l \in [s_t^{-(g_l^+)^2}, s_t^{-(g_l^-)^2}]$ and use that every term of (4) is monotone in each argument on the relevant ranges. Outward rounding uses `nextafter`; exp and log are widened by 4 ulp. For Theorem 4(a,b) we repeat the computation with `mpmath` intervals at 160 bits; all eight proofs run in about 7 seconds.

Certificates. At checkpoints T we try to classify every gap from its enclosure at time T (offset $s_T = t_0 + T$).

- *C1, persistence.* A gap with lower bound g_T^- persists if (5) holds at offset s_T for some $\delta \in (1, g_T^-)$, with $\lambda = 2$, $P = 1$ for chains. This is valid for chains by Lemma 2: the gap dominates the isolated trajectory, which C1 keeps above δ.

- *C2, merging block.* Take a maximal run of consecutive gaps not certified by C1. It forms a block of k layers with diameter $D_B \le \sum g_l^+$, and its outer neighbours persist at levels $\ge \delta$ by C1. Each boundary layer is pulled outwards by at most $g\, s_t^{-g^2} \le \delta s_t^{-\delta^2}$ per step, so the diameter of the block stays below $D_B + 2\delta \sum_{u\ge 0}(s_T+u)^{-\delta^2}$. If this is ≤ 1, every internal gap stays ≤ 1, has non-summable weight, and closes by Proposition 3(ii). The implementation uses the stronger test $D_B + 2(\delta+1)\sum_u (s_T+u)^{-\delta^2} \le 1/\sqrt{\lceil (k-1)/2 \rceil}$, inherited from an earlier path-product argument; it implies the test above.
- *Two blocks.* $d_T^+ \le 1$ certifies merging (Theorem 1(i)); C1 certifies persistence.

A partition is certified when every gap is classified. Enclosures of $\kappa(t_0)$ are pairs (d^-, d^+) with d^- certified to merge and d^+ certified to persist; by Theorem 1(ii) $\kappa \in (d^-, d^+]$.

t_0	validated enclosure of $\kappa(t_0)$	lower bound	upper bound	$1 + \frac{\log\log t_0}{\log t_0}$
3	[1.81765, 1.81781]	1.4024	2.1567	1.0856
5	[1.66856, 1.66879]	1.3742	1.8348	1.2957
8	[1.58233, 1.58267]	1.3494	1.6993	1.3521
11	[1.53858, 1.53909]	1.3336	1.6379	1.3647
17	[1.49084, 1.49159]	1.3138	1.5747	1.3676
30	[1.44246, 1.44360]	1.2910	1.5135	1.3599
100	[1.37027, 1.37281]	1.2522	1.4261	1.3316
300	[1.32457, 1.32837]	1.2256	1.3726	1.3053
10^3	[1.28664, 1.29518]	1.2030	1.3296	1.2798
10^4	[1.23745, 1.24598]	1.1719	1.2728	1.2411
10^5	[1.19909, 1.21827]	1.1502	1.2346	1.2122
10^6	[1.16258, 1.20573]	1.1341	1.2069	1.1901

Table 2: Validated enclosures of $\kappa(t_0) = \kappa(t_0; 1, 1)$, the explicit bounds of Theorem 1(vi) and the leading-order term of (v). The enclosures widen at large t_0 because certifying persistence just above κ needs many steps.

ρ	$t_0 = 11$	$t_0 = 30$	$t_0 = 100$	$t_0 = 300$	$t_0 = 10^3$
0.1	[1.3006, 1.3093]	[1.2574, 1.2661]	[1.2213, 1.2342]	[1.1963, 1.2156]	[1.1738, 1.2026]
0.25	[1.3919, 1.3945]	[1.3294, 1.3333]	[1.2800, 1.2886]	[1.2473, 1.2559]	[1.2190, 1.2319]
0.5	[1.4644, 1.4656]	[1.3855, 1.3881]	[1.3250, 1.3289]	[1.2859, 1.2917]	[1.2529, 1.2658]
1	[1.5385, 1.5391]	[1.4424, 1.4436]	[1.3702, 1.3729]	[1.3245, 1.3284]	[1.2866, 1.2952]

Table 3: Validated enclosures of $\kappa(t_0; \rho, \rho)$ for symmetric cut ratios (discrete map, at most 10^6 steps). The increase with ρ is Theorem 1(iv); the size of the shift is Corollary 1.

F Experimental details

E1–E3 (chains and two blocks). Enclosures of κ: bisection with float trajectories, followed by certification of the two ends with at most 10^6 steps ($t_0 \le 10^3$) or 10^7 steps. The jump (Figure 2c) uses 521 initial gaps in $[0.9, 2.2]$ at $t_0 = 11$, all certified. The three-layer grid (E2) is the 240×240 grid with step 0.0125; the largest certification time is 172,282 layers. The normalised boundaries of Figure 1b are float estimates: bisection in u_0 on trajectories of length 10^5 (10^6 for $t_0 = 10^5$), for 100 values $v_0 \in [0.02, 2]\,\kappa$ at each t_0. E3 draws 4000 vectors of four gaps uniformly from $[0.2, 2.6]$ at $t_0 = 17$; 3972 are certified (the largest certification time is 222,940 layers), and 5 gaps are excluded as ambiguous for the early-depth rule. Partition sizes: 15 runs end with one cluster, 648 with two, 1980 with three, 1138 with four and 191 with five.

$\lambda = 2\rho$	$L = 11.6$	$L = 83.2$	$L = 884$	$L = 9.4 \cdot 10^3$	$L = 10^5$
0.2, two terms	$+0.429$	$+0.283$	$+0.212$	$+0.170$	$+0.141$
0.2, three terms	-0.030	-0.006	$+0.008$	$+0.011$	$+0.009$
0.5, two terms	$+0.275$	$+0.199$	$+0.165$	$+0.138$	$+0.116$
0.5, three terms	-0.092	-0.037	-0.004	$+0.004$	$+0.005$
1, two terms	$+0.164$	$+0.140$	$+0.132$	$+0.115$	$+0.099$
1, three terms	-0.132	-0.057	-0.012	$+0.000$	$+0.002$
2, two terms	$+0.057$	$+0.083$	$+0.100$	$+0.093$	$+0.082$
2, three terms	-0.168	-0.075	-0.018	-0.003	$+0.000$

Table 4: Residual $(\kappa-1)L-[\log L-\frac{1}{2}\log\log L+\frac{1}{2}\log\lambda]$ of the two-term law (Theorem 1(v), which proves that it tends to 0), and the residual after adding the third term $\frac{\log\log L-\log\lambda+2}{4\log L}$ (Proposition 2, which proves that this residual is $o(1/\log L)$). Computed in the continuum limit $dx/d\tau = -\lambda(1+x)e^{-\tau(2x+x^2)}$, $\tau = \log s$, by shooting for the critical $x(L)$. The continuum values lie inside the discrete enclosures of Table 2 for $100 \le t_0 \le 10^6$.

u_0	v_0	proved limit partition	gap 1	certified at layer T
1.4	0.1	one cluster	merges	72
1.4	0.5	$\{1\},\{2,3\}$	persists	2
1.4	0.95	$\{1\},\{2,3\}$	persists	3
1.4	1.5	$\{1,2\},\{3\}$	merges	15
1.4	3.0	$\{1,2\},\{3\}$	merges	12
1.2	0.95	$\{1\},\{2,3\}$	persists	15

Table 5: Initial conditions of the three-layer chain at $t_0 = 11$ whose limiting partition is proved with `mpmath` interval arithmetic (Theorem 4(b)); T is the layer at which all certificates hold. For comparison, $\kappa(11) \in (1.538, 1.540]$.

E4 (noisy SBMs). SBM with $K = 4$ blocks of 150 nodes, $p_{\text{in}} = 0.2$, $p_{\text{out}} \in \{0, 0.005, 0.01, 0.02, 0.05\}$, 5 seeds each. Features in $\mathbb{R}^8$: block k has mean $\frac{3}{\sqrt{2}}e_k$ (pairwise distance 3) plus $\mathcal{N}(0, \sigma^2 I)$ noise, $\sigma \in \{0, 0.03, 0.06, 0.1, 0.15, 0.2, 0.25, 0.3, 0.4, 0.5\}$. We run the full node-level dynamics for $T = 4000$ layers with $t_0 = 602$. The final partition is the set of connected components of edges shorter than 0.05. We also record the fraction of edges with lengths in $[0.05, 0.9]$ (unresolved, by Proposition 3) and the ARI between the partitions at layers 2000 and 4000. At most $3.3 \cdot 10^{-4}$ of the edges are unresolved in any run, and this ARI is at least 0.92 in every run. The threshold-graph rule uses $\kappa(t_0 + t_1)$ for $t_1 \in \{0, 5, 20\}$ ($\kappa \approx 1.3015, 1.3012, 1.3005$). The certificates of Theorem 5 are evaluated on the exact initial conditions, scanning $\eta \in (0, (1-D)/2]$ and taking the smallest admissible δ; the hull form lower-bounds $\operatorname{dist}(C_k, C_l)$ by projecting onto the direction joining block means.

E5 (learned metric). SBM with 3 blocks of 120 nodes, $p_{\text{in}} = 0.15$, $p_{\text{out}} = 0.01$, $t_0 = 362$, $\kappa \approx 1.318$. Features in $\mathbb{R}^8$: class means on a triangle with side 1 in coordinates 1–2, a binary nuisance factor (independent of the class) with value 0 or 3 in coordinate 3, and $\mathcal{N}(0, 0.15^2 I)$ noise in all coordinates. Metrics: $W = I$; the projection onto coordinates 3–4 (nuisance); and the LDA projection fitted on the 20% training nodes, rescaled so that the closest pair of training class means is 2κ apart (calibrated) or 0.4κ apart (too small). Attention uses $z = Wh$ and values h. A logistic-regression probe is trained on h at each depth on the training nodes and evaluated on the rest.

E9 (learned attention). *Two blocks:* the margin map (7) with $\rho_A = \rho_B = 1$ in float64. A trajectory is classified as merging once $\min(M_A, M_B) \le 1$ (Theorem 6(i)) and as persisting once certificate C1′ holds; C1′ is checked every 64 layers with $\delta^2 = 1 + f\,(m-1)$, $f \in$

$\{0.2, 0.4, 0.6, 0.8\}$. Trajectories unresolved after $2 \cdot 10^6$ layers (critical slowing down, at most 24 of the 36 bisection steps per point) are counted as persisting. The diagonal values are 2.3674 at $t_0 = 11$, i.e. $\kappa = 1.5386$, inside the enclosure of Theorem 4(a), and 1.6574 at $t_0 = 10^3$. *Graphs:* the SBMs of E4 with $p_{\text{out}} = 0.01$ and $\sigma \in \{0.1, 0.2, 0.3\}$, 5 seeds each, with all features shifted by $v = \nu e_1$ towards the mean of block 1. The score is $f(x, y) = x^\top J y$ with $J = 2I$ (tied queries and keys) or $J = 2I + 0.3\, G/\|G\|_2$ with standard Gaussian G (untied; $\lambda_{\min}(J_s) \geq 1.7$). With $J_s = 2I$ the symmetric part of every margin is the squared distance of E4. At the block means, the margin of every other block towards block 1 is $9 - 3\sqrt{2}\,\nu$, which crosses $\kappa(t_0)^2$ at $\nu \approx 1.72$; the margins in the opposite direction are $9 + 3\sqrt{2}\,\nu$. We run 4000 layers at $t_0 = 602$ and read off the final partition as in E4. The rules are evaluated after $t_1 \in \{0, 5, 20\}$ layers with threshold $\kappa(t_0 + t_1)^2$.

E8 (real graphs). *Data:* Roman-empire and Amazon-ratings with their ten official 50/25/25 splits [27] (edges symmetrised, self-loops added), and Peptides-func with its official split [8]; sources and SHA-256 hashes are listed with the code. *Untrained dynamics:* the values h are the first 64 principal components of the standardised node features (of the one-hot atom features for Peptides-func). The PCA metric keeps the leading 16 components (8 for Peptides-func), scaled so that the median non-self edge has length $s\,\kappa(t_0)$, $s \in \{0.5, 1, 2\}$. The LDA metric is fitted on the training nodes of split 0 and scaled so that the closest pair of class means is 2κ or 0.4κ apart. We run 4000 layers at $t_0 = |V| + 2$ ($t_0 = 152$ for the molecules) and read off the final partition and the probe accuracies as in E4 and E5. *Training:* on the node graphs all models have width 64, weight decay $5 \cdot 10^{-4}$ and dropout 0.2, and train with full-batch Adam for at most 150 epochs with patience 30 on validation accuracy. On split 0 each (model, depth) picks the learning rate in $\{0.01, 0.003\}$ by validation accuracy; splits 1 and 2 reuse it. The Gaussian networks use the default temperature start ($m = 1$). For Peptides-func the node states are mean-pooled into a 10-label readout, trained for 30 epochs (batch 128, learning rate 10^{-3} with a cosine schedule, no weight decay, dropout 0.1, one seed) and scored by average precision. *Trained networks:* every saved cooled Gaussian network (both learning rates on split 0, and splits 1 and 2) is rerun from its input states $Wx/\sqrt{c}$ for 4000 layers at $t_0 = 11$; the threshold-graph rule uses $\kappa(11 + t_1)$ with $t_1 \in \{0, K\}$.

E6b/E7 (training): data. The CSBMs have 3 classes of 150 nodes and features in $\mathbb{R}^{16}$: class means in random directions with norm $\mu/\sqrt{2}$, plus $\mathcal{N}(0, \sigma^2 I)$ noise. Homophilous clean: $p_{\text{in}} = 0.05$, $p_{\text{out}} = 0.005$, $\mu = 3$, $\sigma = 0.25$; homophilous noisy: the same graph with $\mu = 1.5$, $\sigma = 1$; heterophilous: $p_{\text{in}} = 0.005$, $p_{\text{out}} = 0.04$, $\mu = 3$, $\sigma = 1$. The ring task has 3 classes, each a ring lattice of 200 nodes joined to the neighbours at ring distance 1 and 2. Each node receives, with probability p_x (0.05 sparse, 0.2 dense), one edge to a uniformly chosen node of another class. Four anchor nodes per class (2%) carry features $\mu_y + \mathcal{N}(0, 0.3^2 I)$ with $\|\mu_y\| = 3$ in $\mathbb{R}^{16}$. All other nodes carry pure noise $\mathcal{N}(0, I)$, so a typical node is about 12 hops from the nearest anchor of its class. Every graph has self-loops. Splits are random 20/20/60 (train/validation/test), and seed s draws its own graph, features and split.

E6b/E7 (training): models. Every model maps the input with a linear layer to width 16 and classifies the final node states with a linear readout. GCN, residual GCN ($h \leftarrow h + \text{ReLU}(\hat{A} h W_l)$) and PairNorm (scale s) use a weight matrix per layer. GCNII uses $\alpha \in \{0.1, 0.3\}$ and $\lambda = 0.5$. APPNP ($\alpha \in \{0.1, 0.3\}$) and SGC use the input map only. Linear GRAND (step size $\tau \in \{0.5, 0.1\}$) adds query and key maps for one dot-product attention, computed from the input states and reused at every step. GAT and residual GAT use one head of width 16. The Gaussian networks apply K layers of (1) to $z = Wx$, and the attention is computed and applied on the same states. The cooled version uses temperature $c/\log(l + 11)$ at layer l, and the constant one uses T. The parameter c (or T) is learned in log scale and initialised so that the first-layer temperature equals m times the median squared edge length of Wx at initialisation, with $m \in \{1, 1/30\}$. The cooled dot-product network replaces the Gaussian score by $\langle W_Q z_i, W_K z_j \rangle$, so its score is $f(x, y) = x^\top J y$ with

$J = W_Q^\top W_K$ and $b = 0$ (Section 7). The maps $W_Q, W_K \in \mathbb{R}^{16\times 16}$ are shared by all layers and start at the identity, and c is initialised in the same way, with the median absolute margin $|m_{ij}|$ over the edges in place of the squared edge length. Training is full-batch Adam for at most 300 epochs, with patience 60 on validation accuracy and the epoch of best validation accuracy reported. Each (task, model, depth, seed) picks the best of 8 configurations on validation accuracy: learning rate $\{0.01, 0.003\}$, dropout $\{0, 0.5\}$, and the two values of the model-specific hyperparameter above. Models without one use weight decay $\{5 \cdot 10^{-4}, 0\}$ instead; all others use $5 \cdot 10^{-4}$. Tables 6 and 7 give all results (5 seeds, mean and 95% confidence half-width). The 1675 tasks of 8 trainings each took under 12 hours of wall time in total on an 8-core laptop CPU, and the 300 tasks of the dot-product network another 3.6 hours. For the dot-product network, the run reported for each (task, depth, seed) is rerun in float64 from its input states $z = Wx$ divided by $\sqrt{c}$, so that margins are in units of c, with its learned J and $t_0 = 11$ for 4000 layers. Test accuracy is read with the trained readout at layers K and 4000. The final partition is read off as in E4, and a partition counts as resolved if at most 1% of the edges have lengths between 10^{-6} and 10^{-2} times the median initial edge length and the ARI between the partitions at layers 2000 and 4000 exceeds 0.99. The rules use threshold $\kappa(11 + t_1)^2$, with $\kappa(11) = 1.5386$, $\kappa(16) = 1.4968$ and $\kappa(31) = 1.4400$.

model	$K=2$	$K=8$	$K=16$	$K=32$	$K=64$	$K=128$
heterophilous						
MLP ($K=0$)			.850 ±.026			
GCN	.886 ±.014	.332 ±.021	.324 ±.007	.324 ±.018	.325 ±.014	.333 ±.021
GCN + residual	.962 ±.012	**.975** ±.008	**.964** ±.013	**.949** ±.017	.501 ±.237	.341 ±.034
GCN + PairNorm	.893 ±.024	.750 ±.038	.652 ±.074	.593 ±.055	.578 ±.051	.555 ±.046
GCNII	.851 ±.035	.836 ±.032	.810 ±.057	.721 ±.044	.723 ±.036	.720 ±.048
APPNP	.841 ±.018	.728 ±.023	.729 ±.023	.729 ±.023	.729 ±.023	.729 ±.023
SGC	.883 ±.015	.350 ±.051	.324 ±.012	.324 ±.012	.324 ±.012	.324 ±.012
GAT	.885 ±.014	.365 ±.086	.322 ±.015	.322 ±.015	.322 ±.015	.322 ±.015
GAT + residual	**.966** ±.014	.960 ±.024	.961 ±.012	.927 ±.037	**.913** ±.026	.320 ±.016
GRAND (linear)	.811 ±.027	.714 ±.038	.649 ±.054	.581 ±.057	.519 ±.063	.325 ±.016
Gaussian, constant T	.850 ±.026	.850 ±.024	.852 ±.029	.851 ±.028	.851 ±.027	.842 ±.024
Gaussian, cooled	.850 ±.026	.849 ±.026	.847 ±.026	.850 ±.030	.845 ±.027	**.849** ±.027
dot-product, cooled	.821 ±.046	.676 ±.049	.618 ±.088	.620 ±.076	.635 ±.077	.599 ±.048
homophilous, clean						
MLP ($K=0$)			.994 ±.010			
GCN	.996 ±.002	.947 ±.051	.494 ±.227	.311 ±.014	.327 ±.012	.312 ±.014
GCN + residual	.998 ±.002	.984 ±.020	.990 ±.010	.990 ±.004	.971 ±.021	.578 ±.238
GCN + PairNorm	.992 ±.004	.982 ±.006	.984 ±.007	.982 ±.012	.981 ±.011	.985 ±.005
GCNII	.994 ±.004	.993 ±.004	.995 ±.003	.996 ±.003	.996 ±.003	.996 ±.003
APPNP	.997 ±.001	.998 ±.002	.997 ±.001	.997 ±.001	.997 ±.001	.997 ±.001
SGC	.995 ±.004	.982 ±.007	.983 ±.004	.420 ±.200	.313 ±.023	.313 ±.023
GAT	.991 ±.004	.981 ±.010	.775 ±.121	.322 ±.015	.322 ±.015	.322 ±.015
GAT + residual	.995 ±.008	.993 ±.004	.995 ±.003	.993 ±.004	.980 ±.011	.448 ±.043
GRAND (linear)	**.999** ±.001	.999 ±.002	.998 ±.002	.996 ±.004	.996 ±.004	.997 ±.001
Gaussian, constant T	.991 ±.004	**.999** ±.001	.999 ±.002	**.999** ±.001	.997 ±.003	**1.000** ±.000
Gaussian, cooled	.992 ±.004	**.999** ±.001	**.999** ±.001	**.999** ±.001	**.999** ±.001	.999 ±.001
dot-product, cooled	.996 ±.004	.998 ±.004	.997 ±.003	.998 ±.003	.995 ±.004	.998 ±.003
homophilous, noisy						
MLP ($K=0$)			.573 ±.045			
GCN	.981 ±.006	.971 ±.014	.407 ±.187	.311 ±.014	.327 ±.012	.312 ±.014
GCN + residual	.950 ±.031	.986 ±.005	.986 ±.006	.976 ±.020	.616 ±.281	.425 ±.134
GCN + PairNorm	.970 ±.015	.981 ±.007	.985 ±.009	**.988** ±.008	**.983** ±.007	**.983** ±.007
GCNII	.966 ±.014	.980 ±.010	.962 ±.022	.948 ±.020	.938 ±.030	.941 ±.014
APPNP	.953 ±.022	.942 ±.030	.947 ±.023	.943 ±.018	.943 ±.018	.943 ±.018
SGC	**.981** ±.012	.982 ±.005	.979 ±.007	.396 ±.152	.326 ±.024	.326 ±.024
GAT	.970 ±.012	.977 ±.010	.721 ±.071	.322 ±.015	.322 ±.015	.322 ±.015
GAT + residual	.943 ±.022	**.987** ±.007	**.989** ±.007	.978 ±.012	.964 ±.013	.461 ±.088
GRAND (linear)	.895 ±.030	.976 ±.012	.981 ±.006	.962 ±.016	.979 ±.012	.973 ±.006
Gaussian, constant T	.956 ±.027	.983 ±.010	.967 ±.010	.689 ±.187	.571 ±.049	.567 ±.036
Gaussian, cooled	.956 ±.025	.984 ±.009	.961 ±.012	.693 ±.188	.573 ±.045	.571 ±.047
dot-product, cooled	.967 ±.014	.960 ±.016	.897 ±.063	.635 ±.118	.587 ±.061	.605 ±.059

Table 6: Test accuracy on the CSBM regimes (E6b): mean ± 95% confidence half-width over 5 seeds, each tuned over 8 configurations. Bold marks the best mean in each column. The MLP row is the depth-0 baseline.

model	$K=2$	$K=8$	$K=16$	$K=32$	$K=64$	$K=128$
sparse cross-edges ($p_x = 0.05$)						
MLP ($K=0$)			.324 ±.030			
GCN	.693 ±.019	.882 ±.045	.388 ±.143	.318 ±.011	.316 ±.008	.322 ±.009
GCN + residual	.529 ±.027	.768 ±.052	.796 ±.042	.853 ±.030	.413 ±.194	.337 ±.028
GCN + PairNorm	**.747** ±.032	**.894** ±.025	.913 ±.016	**.939** ±.030	.857 ±.085	.645 ±.073
GCNII	.711 ±.035	.877 ±.047	**.929** ±.037	.892 ±.036	.778 ±.034	.791 ±.031
APPNP	.458 ±.043	.522 ±.035	.534 ±.026	.536 ±.030	.537 ±.030	.537 ±.030
SGC	.471 ±.032	.621 ±.030	.693 ±.018	.780 ±.039	.876 ±.036	.826 ±.107
GAT	.701 ±.049	.848 ±.029	.760 ±.072	.654 ±.084	.319 ±.009	.319 ±.009
GAT + residual	.541 ±.032	.749 ±.056	.809 ±.042	.831 ±.050	.763 ±.060	.357 ±.023
GRAND (linear)	.444 ±.031	.683 ±.044	.778 ±.029	.807 ±.049	**.878** ±.020	**.894** ±.019
Gaussian, constant T	.442 ±.035	.644 ±.047	.754 ±.043	.810 ±.037	.779 ±.073	.759 ±.070
Gaussian, cooled	.451 ±.029	.638 ±.047	.759 ±.041	.783 ±.048	.750 ±.069	.662 ±.029
dot-product, cooled	.513 ±.040	.781 ±.023	.801 ±.079	.779 ±.089	.734 ±.070	.697 ±.036
dense cross-edges ($p_x = 0.2$)						
MLP ($K=0$)			.332 ±.014			
GCN	.672 ±.037	.758 ±.034	.365 ±.078	.334 ±.013	.334 ±.019	.329 ±.017
GCN + residual	.491 ±.030	.731 ±.051	.772 ±.032	.809 ±.023	.403 ±.120	.323 ±.011
GCN + PairNorm	**.732** ±.012	**.878** ±.043	**.913** ±.021	**.876** ±.047	.748 ±.043	.742 ±.124
GCNII	.691 ±.041	.843 ±.055	.856 ±.064	.688 ±.078	.686 ±.070	.679 ±.074
APPNP	.430 ±.055	.474 ±.039	.464 ±.015	.457 ±.020	.457 ±.019	.457 ±.019
SGC	.481 ±.043	.543 ±.030	.562 ±.031	.611 ±.012	.439 ±.102	.345 ±.036
GAT	.633 ±.021	.761 ±.085	.584 ±.055	.352 ±.043	.324 ±.012	.324 ±.012
GAT + residual	.536 ±.037	.706 ±.052	.716 ±.043	.785 ±.018	.706 ±.069	.333 ±.022
GRAND (linear)	.449 ±.024	.619 ±.040	.659 ±.030	.709 ±.056	**.759** ±.048	**.756** ±.027
Gaussian, constant T	.407 ±.048	.562 ±.044	.626 ±.063	.629 ±.045	.486 ±.083	.372 ±.090
Gaussian, cooled	.405 ±.048	.551 ±.039	.571 ±.057	.598 ±.045	.401 ±.064	.341 ±.035
dot-product, cooled	.491 ±.035	.679 ±.043	.692 ±.060	.598 ±.077	.456 ±.015	.359 ±.040

Table 7: Test accuracy on the long-range ring task (E7), as in Table 6. Chance is 1/3.

model	$K=2$	$K=4$	$K=8$	$K=16$	$K=32$
Roman-empire					
MLP ($K=0$)			.644 ±.001		
GCN	.481 ±.007	.260 ±.007	.149 ±.018	.161 ±.019	.168 ±.016
GCN + residual	**.775** ±.001	.786 ±.006	.786 ±.008	**.775** ±.011	.603 ±.023
GCN + PairNorm	.566 ±.013	.521 ±.004	.412 ±.011	.329 ±.008	.266 ±.030
GCNII	.662 ±.010	.658 ±.004	.637 ±.004	.613 ±.010	.555 ±.019
APPNP	.549 ±.005	.510 ±.003	.501 ±.003	.505 ±.006	.508 ±.005
SGC	.426 ±.006	.302 ±.007	.246 ±.009	.208 ±.004	.190 ±.007
GAT	.460 ±.013	.364 ±.021	.169 ±.005	.129 ±.019	.129 ±.019
GAT + residual	.767 ±.004	**.800** ±.005	**.803** ±.010	.753 ±.024	.633 ±.062
GRAND (linear)	.670 ±.011	.623 ±.008	.577 ±.007	.553 ±.006	.538 ±.004
Gaussian, constant T	.643 ±.004	.643 ±.003	.642 ±.002	.641 ±.003	.643 ±.002
Gaussian, cooled	.642 ±.002	.642 ±.001	.642 ±.001	.642 ±.003	**.643** ±.002
Amazon-ratings					
MLP ($K=0$)			.386 ±.004		
GCN	.426 ±.007	.368 ±.000	.368 ±.000	.368 ±.000	.368 ±.000
GCN + residual	.424 ±.005	.430 ±.004	.439 ±.005	.443 ±.015	.382 ±.028
GCN + PairNorm	**.470** ±.003	**.466** ±.004	**.463** ±.005	**.452** ±.002	.431 ±.034
GCNII	.413 ±.044	.413 ±.044	.407 ±.038	.426 ±.002	.420 ±.003
APPNP	.423 ±.004	.424 ±.006	.424 ±.003	.424 ±.003	.424 ±.003
SGC	.423 ±.003	.424 ±.001	.422 ±.003	.425 ±.004	.424 ±.003
GAT	.431 ±.007	.432 ±.010	.402 ±.033	.335 ±.065	.335 ±.065
GAT + residual	.425 ±.002	.428 ±.001	.444 ±.007	.442 ±.005	**.434** ±.003
GRAND (linear)	.415 ±.004	.424 ±.002	.430 ±.003	.430 ±.002	.426 ±.004
Gaussian, constant T	.415 ±.004	.417 ±.005	.419 ±.004	.419 ±.007	.385 ±.033
Gaussian, cooled	.415 ±.001	.417 ±.003	.419 ±.006	.417 ±.011	.395 ±.016

Table 8: Test accuracy on the real node graphs (E8): mean ± 1.96 standard errors over three official splits. Split 0 selects the learning rate per (model, depth) on validation accuracy, and splits 1 and 2 reuse it. Bold marks the best mean in each column.

model	$K = 2$	$K = 8$	$K = 32$
Peptides-func (average precision, one seed)			
MLP ($K = 0$)		.253	
GCN	.349	.337	.169
GCN + residual	.346	.378	.268
GCNII	.307	.310	.281
GAT + residual	**.373**	**.421**	**.379**
Gaussian, constant T	.253	.247	.247
Gaussian, cooled	.254	.246	.247

Table 9: Test average precision on Peptides-func (E8), one seed, 30 epochs, protocol in Section F. Bold marks the best in each column.